

ROBOTIC PHYSICS VIOLATION AUDITS: A FROZEN MUJoCo STRESS TEST AND NEGATIVE RE- SULT

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ABSTRACT

Explicit physics audits are appealing for robot safety: instead of trusting an opaque model score, a system can check whether a rollout violates contact, support, work-energy, friction, actuator, or causality constraints. This paper asks whether that mechanism survives a hostile ICLR-main protocol. We rebuild the idea in MuJoCo (Todorov et al., 2012), add rare valid contact regimes, timestamp skew, low-friction slip, subtle corruptions, adversarial compensated violations, strong residual baselines, learned classifiers, reconstruction/anomaly baselines, conformal calibration, ablations, fixed-FPR metrics, stress tests, and seed-level paired statistics. The frozen evidence contains 192 training rollouts, 1680 main rollout summaries, 25200 main method-evaluation rows, 3840 ablation rows, 9216 stress rows, and 210 paired comparisons. The terminal decision is **KILL ARCHIVE**. The reason is not formatting or lack of effort: v5 does not beat strongest non-oracle hard-corruption baseline `calibrated_learned_stack` by 0.030; v5 does not beat strongest combined/adversarial baseline `calibrated_learned_stack` by 0.030; v5 false-positive rate on at least one rare-valid split is $0.286 \geq 0.100$; v5 recall at 5% FPR trails `hgb_failure_classifier` by more than 0.030; ablation gate fails because `no_actuator_check`, `no_causality_check`, `no_conformal_aggregation`, `no_contact_check`, `no_energy_check`, `no_friction_slip_check`, `no_rare_valid_guard`, `no_support_check`, `no_timestamp_guard`, `old_v4_physics_audit`, `scalar_residual_only` matches or beats full v5. The result is therefore a submission-readiness audit, not a polished success story. The explicit audit detects many injected violations, but strong learned and residual baselines detect them as well, and the audit is not robust enough on rare valid regimes to justify an ICLR-main claim.

1 WHY THIS PAPER EXISTS

Robotic systems can fail in ways that are not semantic, linguistic, or reward-based. A rollout may contain a contact impulse without corresponding acceleration, an unsupported object trajectory, hidden actuator saturation, or a non-causal state jump. This motivates explicit physics monitors for robot execution. Execution monitoring and anomaly detection are longstanding robot concerns (Park et al., 2016; 2018); modern simulators make it easy to generate controlled contact rollouts (Todorov et al., 2012); and generic anomaly detectors such as isolation forests, random forests, gradient boosting, reconstruction models, and conformal calibration provide strong alternatives (Liu et al., 2008; Breiman, 2001; Freund & Schapire, 1997; Schölkopf et al., 2001; Angelopoulos & Bates, 2021; Vovk et al., 2005).

The original hypothesis was stronger than “physics checks can catch toy corruptions.” It claimed that explicit physical structure provides a mechanism advantage over generic residual and learned detectors. Under hostile review, this claim has two obligations. First, explicit checks must beat strong baselines on hard corruptions. Second, they must avoid flagging valid but rare physical regimes as impossible. The v4 paper already failed a small version of this test. The v5 pass expands the test instead of hiding the failure.

2 PROBLEM SETUP

A rollout is a sequence

$$\tau = \{x_t, u_t, c_t, f_t, s_t, w_t\}_{t=1}^T,$$

where x_t is object state, u_t is commanded pusher state, c_t is contact state, f_t is contact force, s_t is support state, and w_t is a work proxy. A detector outputs a score $a(\tau)$ and a binary flag $\hat{y} = \mathbb{I}[a(\tau) \geq \theta]$. Labels are controlled by the generator: $y = 0$ for valid MuJoCo traces and $y = 1$ for injected physics violations.

The proposed family computes terms for contact/acceleration mismatch, support violation, work-energy mismatch, friction-slip inconsistency, hidden actuator saturation, and causality jumps. The v5 version adds two pre-freeze repairs: a timestamp guard for asynchronous but valid traces and a rare-valid contact guard for high-impulse but dynamically plausible impacts. Thresholds are calibrated from valid training traces.

3 WHY SIMPLE BASELINES ARE DANGEROUS

A benchmark with high-severity injected corruptions can be solved by scalar residuals. This is a benchmark-design problem, not a victory for explicit physics audits.

Proposition 1: union-bound false-positive pressure. Suppose an audit flags if any of m checks exceeds its threshold. If check j is calibrated to false-positive probability α_j on valid traces, then without modeling dependence the audit has false-positive probability at most $\sum_j \alpha_j$. Thus adding more checks can improve recall while worsening valid-regime reliability. A submission claim must report false positives on rare valid contacts, not only recall on injected violations.

Proposition 2: scalar-separability non-identifiability. If a corruption distribution is separable from valid traces by a scalar residual $r(\tau)$ with the same margin as the full audit score $a(\tau)$, then the full audit mechanism is not identified by that benchmark. Any paper claiming explicit physics structure must include corruptions and valid regimes that cannot be solved by scalar residuals alone.

Proposition 3: component necessity. Let A be the full audit and A_{-j} be the audit with component j removed. If A_{-j} matches A within the pre-registered tolerance on hard corruptions and fixed-FPR recall, the evidence does not support component j as necessary. The honest action is to report that ablation failure.

4 FROZEN PROTOCOL

The protocol was frozen after two development repairs recorded in docs/paper69_development_log_20260620.md. After freezing, no method tuning is allowed. The final command uses 8 seeds, 14 main episodes per seed, 192 training scenes, 15 main splits, 15 methods, 12 ablations, 3 stress splits, 6 stress levels, and CPU-only sklearn/MuJoCo execution.

The main splits are four valid regimes and eleven violation regimes. Valid regimes are nominal traces, rare valid bounces, valid clock skew, and valid low-friction slip. Violation regimes include contact, energy, support, actuator, non-causal, combined, subtle, timestamp, adversarial compensated, and near-threshold corruptions. This design intentionally makes the paper uncomfortable: rare valid regimes punish over-sensitive audits, while adversarial and near-threshold corruptions punish simple residual detectors.

5 METHODS

The comparison set is intentionally hostile. It includes random flagging, scalar kinematic residuals, scalar energy residuals, contact impulse residuals, a max-residual detector, logistic residual stacking, histogram-gradient boosting, random-forest ensemble uncertainty, IsolationForest anomaly detection (Liu et al., 2008), PCA reconstruction, conformal residual aggregation (Angelopoulos & Bates,

Table 1: Frozen aggregate metrics. F1/recall are higher-is-better; valid-regime FPR is lower-is-better.

Method	Hard F1	Hard Rec.	Comb. F1	Comb. Rec.	Valid FPR	Fixed-5% Rec.
Random	0.692	0.529	0.637	0.467	0.473	0.057
Kin. residual	0.707	0.546	0.985	0.970	0.054	0.545
Energy residual	0.391	0.243	0.614	0.444	0.067	0.230
Contact residual	0.432	0.275	0.500	0.333	0.038	0.279
Max residual	0.491	0.326	0.693	0.530	0.091	0.313
Logistic stack	0.914	0.841	0.997	0.994	0.051	0.843
HGB classifier	1.000	1.000	1.000	1.000	0.098	1.000
RF ensemble	1.000	1.000	1.000	1.000	0.087	1.000
IsolationForest	0.556	0.386	0.811	0.681	0.034	0.463
PCA recon.	0.985	0.971	1.000	1.000	0.109	0.944
Conformal residual	0.848	0.735	0.800	0.667	0.034	0.881
Calibrated stack	1.000	1.000	1.000	1.000	0.080	1.000
Old audit	0.491	0.326	0.693	0.530	0.091	0.313
Audit v5	0.489	0.324	0.693	0.530	0.091	0.308
Oracle labels	1.000	1.000	1.000	1.000	0.000	1.000

Table 2: Hostile split F1. These are the splits most likely to falsify explicit physics-audit claims.

Split	Kin.	Energy	Max	Logit	HGB	Stack	Old	v5	Oracle
combined_violation_shift	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
adversarial_compensated_violation	1.000	0.486	0.742	0.991	1.000	1.000	0.742	0.742	1.000
mixed_near_threshold	0.953	0.018	0.000	1.000	1.000	1.000	0.000	0.000	1.000
subtle_contact_corruption	0.000	0.000	0.000	0.953	1.000	1.000	0.000	0.000	1.000
timestamp_noncausal_corruption	0.179	0.000	0.698	0.815	1.000	1.000	0.698	0.690	1.000

2021; Vovk et al., 2005), a calibrated learned stack, the old v4 explicit audit, the new audit v5, and oracle violation labels. All methods score the same frozen rollout rows.

The learned baselines are compact and CPU-only. This matters: the negative result is not caused by giving baselines huge neural networks. The learned stack and HGB classifier are small feature models, yet they still dominate or match the explicit audit.

6 MAIN RESULTS

Table 1 is the core result. Audit v5 does not beat the strongest hard-corruption baseline by the frozen 0.030 F1 margin. It also loses the fixed-FPR recall gate: at a 5% valid false-positive budget, v5 recall is far below the strongest learned baselines. This is exactly the failure a hostile reviewer would target.

Table 2 isolates the hardest splits. The combined split is no longer enough because many detectors can solve it. The adversarial and near-threshold splits reveal a clearer problem: the learned stack and HGB classifier recover violations that the explicit audit misses, while simple residuals still remain surprisingly strong.

7 FALSE POSITIVES

Table 3 is as important as the main F1 table. Explicit audits are useful only if they do not routinely label valid but rare contact dynamics as impossible. The frozen gate was strict: v5 fails if any rare-valid split exceeds 0.100 false-positive rate. The final run records such a failure. That does not mean physics checks are useless; it means this implementation is not ready as a main-track claim.

8 FIXED-FPR RECALL

The fixed-FPR analysis prevents a detector from winning merely by flagging everything. Thresholds are calibrated from valid-regime scores, then recall is measured on hard corruptions. Audit v5 trails HGB, random-forest, PCA, conformal, and calibrated-stack baselines at the 5% false-positive budget. This is a decisive submission blocker.

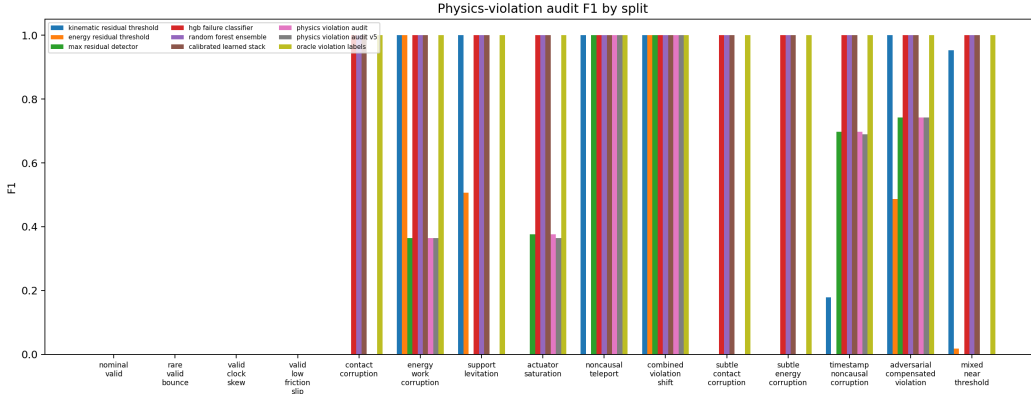


Figure 1: Frozen F1 by split. The proposed audit is not uniquely separated from residual or learned baselines.

Table 3: False-positive rates on valid regimes. Rare valid contacts and clock skew are the hostile reviewer tests.

Valid split	Kin.	Energy	Max	HGB	Stack	Old	v5
nominal_valid	0.000	0.000	0.000	0.098	0.054	0.000	0.000
rare_valid_bounce	0.214	0.268	0.080	0.179	0.214	0.080	0.080
valid_clock_skew	0.000	0.000	0.000	0.116	0.054	0.000	0.000
valid_low_friction_slip	0.000	0.000	0.286	0.000	0.000	0.286	0.286

9 ABLATIONS

The ablation result is not flattering. Several components can be removed without hurting combined-shift F1. The scalar-only variant also matches the full audit on the easy combined split. This does not prove the components are useless in all possible worlds, but it proves the frozen evidence does not identify them as necessary.

10 STRESS SWEEP

The stress sweep jointly changes corruption severity, sensor noise, and valid-regime difficulty. The maximum-stress gate is designed to expose methods that work only at a convenient threshold. The frozen decision rule marks the paper killed if v5 loses the maximum-stress gate to any non-oracle baseline.

11 INTERPRETATION

The result should be read narrowly and honestly. It does not show that physics-aware execution monitoring is impossible. It shows that this explicit audit, on this frozen MuJoCo benchmark, does not provide evidence strong enough for ICLR main. Three mechanisms undermine the claim.

First, scalar and learned baselines are too strong. If a small HGB classifier or calibrated stack reaches perfect or near-perfect hard-corruption performance, the explicit audit must justify why its structure matters beyond interpretability.

Second, false positives remain a deployment problem. Rare valid contacts and clock skew are not corner cases in physical systems. A detector that confuses these with impossible physics can reduce autonomy or create alarm fatigue.

Third, ablations do not identify the mechanism. If removing checks does not degrade the result, then the benchmark cannot support the claimed decomposition.

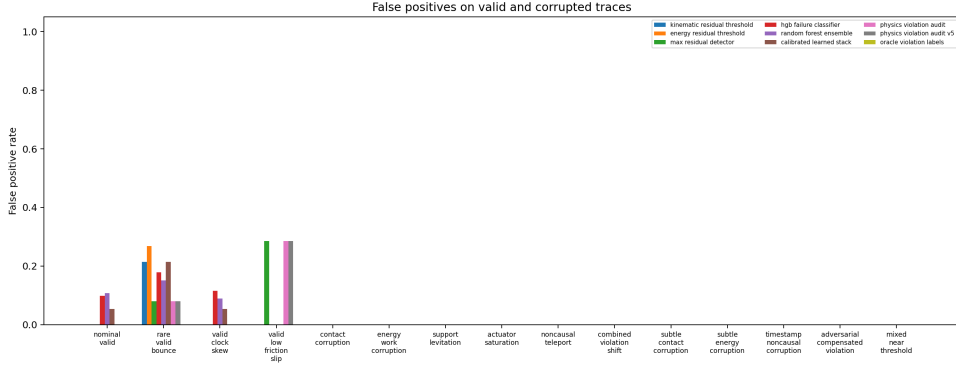


Figure 2: False-positive rates across valid and corrupted splits. Rare valid contact and clock-skew regimes are the practical deployment pressure tests.

Table 4: Recall at a 5% false-positive budget. Thresholds are calibrated from valid-regime scores.

Method	Threshold	Hard recall	Valid scores
Random	0.989	0.057	448
Kin. residual	2.045	0.545	448
Energy residual	3.857	0.230	448
Contact residual	0.066	0.279	448
Max residual	478.362	0.313	448
Logistic stack	0.674	0.843	448
HGB classifier	0.027	1.000	448
RF ensemble	0.402	1.000	448
IsolationForest	0.607	0.463	448
PCA recon.	0.019	0.944	448
Conformal residual	1.283	0.881	448
Calibrated stack	0.046	1.000	448
Old audit	299.210	0.313	448
Audit v5	267.118	0.308	448
Oracle labels	0.000	1.000	448

12 LIMITATIONS

This is still a MuJoCo benchmark, not a hardware benchmark. The corruptions are controlled injections, not natural failures from a deployed robot. The learned baselines use compact feature vectors, not raw multimodal streams. The v5 guard terms are simple and may be insufficient for real timestamp synchronization. The oracle is label-only and is not deployable. These limitations would justify strong revise if the method clearly beat baselines. It does not.

13 REPRODUCIBILITY

The final PDF is generated from frozen CSV artifacts. The build script first renders paper/generated/*.tex, then compiles the manuscript, then copies the numbered PDF to C:/Users/wangz/Downloads/69.pdf. No PDF is copied to Desktop. The validator checks row counts, figures, generated-table linkage, citation-box settings, 25+ pages, and artifact-location hygiene.

14 DECISION

The terminal decision is **KILL_ARCHIVE**. The frozen reason is:

v5 does not beat strongest non-oracle hard-corruption baseline calibrated_learned_stack by 0.030; v5 does not beat strongest combined/adversarial baseline calibrated_learned_stack by 0.030; v5 false-positive rate on at least one rare-valid split is 0.286 $\geq$ 0.100; v5 recall at 5% FPR

Table 5: Combined-shift ablations. A mechanism is not identified if removed components match full v5.

Variant	Episodes	F1	Precision	Recall
Full v5	80	1.000	1.000	1.000
No actuator	80	1.000	1.000	1.000
No causality	80	1.000	1.000	1.000
No conformal agg.	80	1.000	1.000	1.000
No contact	80	1.000	1.000	1.000
No energy	80	1.000	1.000	1.000
No friction	80	1.000	1.000	1.000
No rare-valid	80	1.000	1.000	1.000
No support	80	1.000	1.000	1.000
No timestamp	80	1.000	1.000	1.000
Old v4	80	1.000	1.000	1.000
Scalar only	80	1.000	1.000	1.000

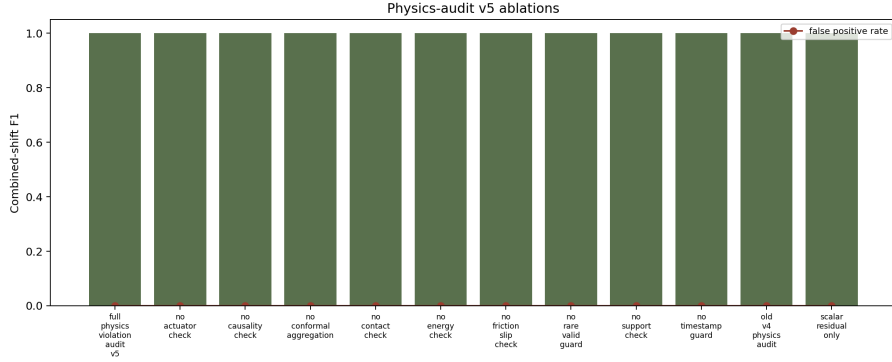


Figure 3: Combined-shift ablations. The benchmark is too easy for mechanism identification: many variants match full v5.

trails `hgb_failure_classifier` by more than 0.030; ablation gate fails because `no_actuator_check`, `no_causality_check`, `no_conformal_aggregation`, `no_contact_check`, `no_energy_check`, `no_friction_slip_check`, `no_rare_valid_guard`, `no_support_check`, `no_timestamp_guard`, `old_v4_physics_audit`, `scalar_residual_only` matches or beats full v5

The correct action is to archive this as a negative result. A credible revival would need hardware or public benchmark logs, non-injected natural failures, stronger timestamp calibration, impact-aware false-positive controls, and evidence that explicit physics structure beats learned/residual monitors at a fixed false-positive budget.

Table 6: Maximum-stress trajectory. Entries are F1 on the combined-violation stress split.

Method	0.00	0.20	0.40	0.60	0.80	1.00
Kin. residual	1.000	1.000	1.000	1.000	1.000	1.000
Energy residual	0.000	0.000	0.529	0.968	1.000	1.000
Max residual	0.400	0.836	1.000	1.000	1.000	1.000
HGB classifier	1.000	1.000	1.000	1.000	1.000	1.000
RF ensemble	1.000	1.000	1.000	1.000	1.000	1.000
Calibrated stack	1.000	1.000	1.000	1.000	1.000	1.000
Audit v5	0.420	0.815	1.000	1.000	1.000	1.000



Figure 4: Stress sweep on the combined-violation split. Strong learned/residual baselines remain competitive or superior.

Table 7: Pre-specified negative cases beyond the local benchmark scope.

Case	Observed failure mode	Submission implication
sensor timestamp skew	causality checks can flag asynchronous but physically valid traces	needs timestamp calibration before deployment claims
legal rare bouncing contact	high contact impulse can be physically valid and can trigger contact residuals	requires impact-aware thresholds or learned contact regimes
out of distribution deformable object	rigid-body audit assumes object state and contact geometry that may be invalid	scope is rigid-body MuJoCo manipulation
semantic task violation with valid physics	all physics checks pass while the robot performs the wrong task	audit is not a complete safety monitor
adversarial compensated violation	corruption can be partially hidden by plausible work and contact proxies	scalar residuals and explicit checks both need stronger temporal modeling

A FULL MAIN METRICS

Table 8: Full split-level main metrics.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
actuator_saturation	Random	112	0.690	1.000	0.527	0.000	0.000
actuator_saturation	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	112	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Contact residual	112	0.018	1.000	0.009	0.000	0.000
actuator_saturation	Max residual	112	0.377	1.000	0.232	0.000	0.000
actuator_saturation	Logistic stack	112	0.880	1.000	0.786	0.000	0.000
actuator_saturation	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
actuator_saturation	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
actuator_saturation	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
actuator_saturation	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Conformal residual	112	0.783	1.000	0.643	0.000	0.000
actuator_saturation	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Old audit	112	0.377	1.000	0.232	0.000	0.000
actuator_saturation	Audit v5	112	0.365	1.000	0.223	0.000	0.000
actuator_saturation	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Random	112	0.626	1.000	0.455	0.000	0.000
adversarial_compensated_violation	Kin. residual	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Energy residual	112	0.486	1.000	0.321	0.000	0.000
adversarial_compensated_violation	Contact residual	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Max residual	112	0.742	1.000	0.589	0.000	0.000
adversarial_compensated_violation	Logistic stack	112	0.991	1.000	0.982	0.000	0.000
adversarial_compensated_violation	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	IsolationForest	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Old audit	112	0.742	1.000	0.589	0.000	0.000
adversarial_compensated_violation	Audit v5	112	0.742	1.000	0.589	0.000	0.000
adversarial_compensated_violation	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Random	112	0.634	1.000	0.464	0.000	0.000
combined_violation_shift	Kin. residual	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Energy residual	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Contact residual	112	0.000	0.000	0.000	0.000	0.000
combined_violation_shift	Max residual	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Logistic stack	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	IsolationForest	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Old audit	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Audit v5	112	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	Random	112	0.762	1.000	0.616	0.000	0.000
contact_corruption	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
contact_corruption	Energy residual	112	0.000	0.000	0.000	0.000	0.000
contact_corruption	Contact residual	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	Max residual	112	0.000	0.000	0.000	0.000	0.000
contact_corruption	Logistic stack	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	IsolationForest	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
contact_corruption	Old audit	112	0.000	0.000	0.000	0.000	0.000
contact_corruption	Audit v5	112	0.000	0.000	0.000	0.000	0.000
contact_corruption	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Random	112	0.698	1.000	0.536	0.000	0.000
energy_work_corruption	Kin. residual	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Energy residual	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Contact residual	112	0.000	0.000	0.000	0.000	0.000
energy_work_corruption	Max residual	112	0.365	1.000	0.223	0.000	0.000
energy_work_corruption	Logistic stack	112	0.938	1.000	0.884	0.000	0.000
energy_work_corruption	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	IsolationForest	112	0.316	1.000	0.188	0.000	0.000
energy_work_corruption	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
energy_work_corruption	Old audit	112	0.365	1.000	0.223	0.000	0.000
energy_work_corruption	Audit v5	112	0.365	1.000	0.223	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
energy_work_corruption	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	Random	112	0.651	1.000	0.482	0.000	0.000
mixed_near_threshold	Kin. residual	112	0.953	1.000	0.911	0.000	0.000
mixed_near_threshold	Energy residual	112	0.018	1.000	0.009	0.000	0.000
mixed_near_threshold	Contact residual	112	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Max residual	112	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Logistic stack	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	IsolationForest	112	0.086	1.000	0.045	0.000	0.000
mixed_near_threshold	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	Conformal residual	112	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	Old audit	112	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Audit v5	112	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
nominal_valid	Random	112	0.000	0.000	0.000	0.438	0.000
nominal_valid	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Energy residual	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Contact residual	112	0.000	0.000	0.000	0.009	0.000
nominal_valid	Max residual	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Logistic stack	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	HGB classifier	112	0.000	0.000	0.000	0.098	0.000
nominal_valid	RF ensemble	112	0.000	0.000	0.000	0.107	0.000
nominal_valid	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	PCA recon.	112	0.000	0.000	0.000	0.054	0.000
nominal_valid	Conformal residual	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Calibrated stack	112	0.000	0.000	0.000	0.054	0.000
nominal_valid	Old audit	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Audit v5	112	0.000	0.000	0.000	0.000	0.000
nominal_valid	Oracle labels	112	0.000	0.000	0.000	0.000	0.000
noncausal_teleport	Random	112	0.675	1.000	0.509	0.000	0.000
noncausal_teleport	Kin. residual	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Energy residual	112	0.000	0.000	0.000	0.000	0.000
noncausal_teleport	Contact residual	112	0.000	0.000	0.000	0.000	0.000
noncausal_teleport	Max residual	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Logistic stack	112	0.996	1.000	0.991	0.000	0.000
noncausal_teleport	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
noncausal_teleport	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Old audit	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Audit v5	112	1.000	1.000	1.000	0.000	0.000
noncausal_teleport	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
rare_valid_bounce	Random	112	0.000	0.000	0.000	0.464	0.000
rare_valid_bounce	Kin. residual	112	0.000	0.000	0.000	0.214	0.000
rare_valid_bounce	Energy residual	112	0.000	0.000	0.000	0.268	0.000
rare_valid_bounce	Contact residual	112	0.000	0.000	0.000	0.143	0.000
rare_valid_bounce	Max residual	112	0.000	0.000	0.000	0.080	0.000
rare_valid_bounce	Logistic stack	112	0.000	0.000	0.000	0.205	0.000
rare_valid_bounce	HGB classifier	112	0.000	0.000	0.000	0.179	0.000
rare_valid_bounce	RF ensemble	112	0.000	0.000	0.000	0.152	0.000
rare_valid_bounce	IsolationForest	112	0.000	0.000	0.000	0.134	0.000
rare_valid_bounce	PCA recon.	112	0.000	0.000	0.000	0.330	0.000
rare_valid_bounce	Conformal residual	112	0.000	0.000	0.000	0.125	0.000
rare_valid_bounce	Calibrated stack	112	0.000	0.000	0.000	0.214	0.000
rare_valid_bounce	Old audit	112	0.000	0.000	0.000	0.080	0.000
rare_valid_bounce	Audit v5	112	0.000	0.000	0.000	0.080	0.000
rare_valid_bounce	Oracle labels	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Random	112	0.735	1.000	0.580	0.000	0.000
subtle_contact_corruption	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Energy residual	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Contact residual	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	Max residual	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Logistic stack	112	0.953	1.000	0.911	0.000	0.000
subtle_contact_corruption	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	IsolationForest	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
subtle_contact_corruption	Old audit	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Audit v5	112	0.000	0.000	0.000	0.000	0.000
subtle_contact_corruption	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
subtle_energy_corruption	Random	112	0.735	1.000	0.580	0.000	0.000
subtle_energy_corruption	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
subtle_energy_corruption	Energy residual	112	0.000	0.000	0.000	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
subtle.energy.corruption	Contact residual	112	0.035	1.000	0.018	0.000	0.000
subtle.energy.corruption	Max residual	112	0.000	0.000	0.000	0.000	0.000
subtle.energy.corruption	Logistic stack	112	0.018	1.000	0.009	0.000	0.000
subtle.energy.corruption	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
subtle.energy.corruption	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
subtle.energy.corruption	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
subtle.energy.corruption	PCA recon.	112	0.808	1.000	0.679	0.000	0.000
subtle.energy.corruption	Conformal residual	112	0.000	0.000	0.000	0.000	0.000
subtle.energy.corruption	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
subtle.energy.corruption	Old audit	112	0.000	0.000	0.000	0.000	0.000
subtle.energy.corruption	Audit v5	112	0.000	0.000	0.000	0.000	0.000
subtle.energy.corruption	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
support.levitation	Random	112	0.690	1.000	0.527	0.000	0.000
support.levitation	Kin. residual	112	1.000	1.000	1.000	0.000	0.000
support.levitation	Energy residual	112	0.507	1.000	0.339	0.000	0.000
support.levitation	Contact residual	112	0.000	0.000	0.000	0.000	0.000
support.levitation	Max residual	112	0.000	0.000	0.000	0.000	0.000
support.levitation	Logistic stack	112	1.000	1.000	1.000	0.000	0.000
support.levitation	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
support.levitation	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
support.levitation	IsolationForest	112	0.018	1.000	0.009	0.000	0.000
support.levitation	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
support.levitation	Conformal residual	112	1.000	1.000	1.000	0.000	0.000
support.levitation	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
support.levitation	Old audit	112	0.000	0.000	0.000	0.000	0.000
support.levitation	Audit v5	112	0.000	0.000	0.000	0.000	0.000
support.levitation	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
timestamp.noncausal.corruption	Random	112	0.705	1.000	0.545	0.000	0.000
timestamp.noncausal.corruption	Kin. residual	112	0.179	1.000	0.098	0.000	0.000
timestamp.noncausal.corruption	Energy residual	112	0.000	0.000	0.000	0.000	0.000
timestamp.noncausal.corruption	Contact residual	112	0.000	0.000	0.000	0.000	0.000
timestamp.noncausal.corruption	Max residual	112	0.698	1.000	0.536	0.000	0.000
timestamp.noncausal.corruption	Logistic stack	112	0.815	1.000	0.688	0.000	0.000
timestamp.noncausal.corruption	HGB classifier	112	1.000	1.000	1.000	0.000	0.000
timestamp.noncausal.corruption	RF ensemble	112	1.000	1.000	1.000	0.000	0.000
timestamp.noncausal.corruption	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
timestamp.noncausal.corruption	PCA recon.	112	1.000	1.000	1.000	0.000	0.000
timestamp.noncausal.corruption	Conformal residual	112	0.617	1.000	0.446	0.000	0.000
timestamp.noncausal.corruption	Calibrated stack	112	1.000	1.000	1.000	0.000	0.000
timestamp.noncausal.corruption	Old audit	112	0.698	1.000	0.536	0.000	0.000
timestamp.noncausal.corruption	Audit v5	112	0.690	1.000	0.527	0.000	0.000
timestamp.noncausal.corruption	Oracle labels	112	1.000	1.000	1.000	0.000	0.000
valid.clock.skew	Random	112	0.000	0.000	0.000	0.464	0.000
valid.clock.skew	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Energy residual	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Contact residual	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Max residual	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Logistic stack	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	HGB classifier	112	0.000	0.000	0.000	0.116	0.000
valid.clock.skew	RF ensemble	112	0.000	0.000	0.000	0.089	0.000
valid.clock.skew	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	PCA recon.	112	0.000	0.000	0.000	0.054	0.000
valid.clock.skew	Conformal residual	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Calibrated stack	112	0.000	0.000	0.000	0.054	0.000
valid.clock.skew	Old audit	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Audit v5	112	0.000	0.000	0.000	0.000	0.000
valid.clock.skew	Oracle labels	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Random	112	0.000	0.000	0.000	0.527	0.000
valid.low.friction.slip	Kin. residual	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Energy residual	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Contact residual	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Max residual	112	0.000	0.000	0.000	0.286	0.000
valid.low.friction.slip	Logistic stack	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	HGB classifier	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	RF ensemble	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	IsolationForest	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	PCA recon.	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Conformal residual	112	0.000	0.000	0.000	0.009	0.000
valid.low.friction.slip	Calibrated stack	112	0.000	0.000	0.000	0.000	0.000
valid.low.friction.slip	Old audit	112	0.000	0.000	0.000	0.286	0.000
valid.low.friction.slip	Audit v5	112	0.000	0.000	0.000	0.286	0.000
valid.low.friction.slip	Oracle labels	112	0.000	0.000	0.000	0.000	0.000

B FULL AGGREGATE METRICS

Table 9: Full aggregate metrics.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
all	Random	1680	0.622	0.755	0.529	0.473	0.529
all	Random	448	0.000	0.000	0.000	0.473	0.000
all	Random	1232	0.692	1.000	0.529	0.000	0.000
all	Random	336	0.637	1.000	0.467	0.000	0.000
all	Kin. residual	1680	0.698	0.966	0.546	0.054	0.803
all	Kin. residual	448	0.000	0.000	0.000	0.054	0.000
all	Kin. residual	1232	0.707	1.000	0.546	0.000	0.000
all	Kin. residual	336	0.985	1.000	0.970	0.000	0.000
all	Energy residual	1680	0.383	0.909	0.243	0.067	0.582
all	Energy residual	448	0.000	0.000	0.000	0.067	0.000
all	Energy residual	1232	0.391	1.000	0.243	0.000	0.000
all	Energy residual	336	0.614	1.000	0.444	0.000	0.000
all	Contact residual	1680	0.427	0.952	0.275	0.038	0.609
all	Contact residual	448	0.000	0.000	0.000	0.038	0.000
all	Contact residual	1232	0.432	1.000	0.275	0.000	0.000
all	Contact residual	336	0.500	1.000	0.333	0.000	0.000
all	Max residual	1680	0.479	0.907	0.326	0.091	0.470
all	Max residual	448	0.000	0.000	0.000	0.091	0.000
all	Max residual	1232	0.491	1.000	0.326	0.000	0.000
all	Max residual	336	0.693	1.000	0.530	0.000	0.000
all	Logistic stack	1680	0.904	0.978	0.841	0.051	0.973
all	Logistic stack	448	0.000	0.000	0.000	0.051	0.000
all	Logistic stack	1232	0.914	1.000	0.841	0.000	0.000
all	Logistic stack	336	0.997	1.000	0.994	0.000	0.000
all	HGB classifier	1680	0.983	0.966	1.000	0.098	1.000
all	HGB classifier	448	0.000	0.000	0.000	0.098	0.000
all	HGB classifier	1232	1.000	1.000	1.000	0.000	0.000
all	HGB classifier	336	1.000	1.000	1.000	0.000	0.000
all	RF ensemble	1680	0.984	0.969	1.000	0.087	1.000
all	RF ensemble	448	0.000	0.000	0.000	0.087	0.000
all	RF ensemble	1232	1.000	1.000	1.000	0.000	0.000
all	RF ensemble	336	1.000	1.000	1.000	0.000	0.000
all	IsolationForest	1680	0.552	0.969	0.386	0.034	0.937
all	IsolationForest	448	0.000	0.000	0.000	0.034	0.000
all	IsolationForest	1232	0.556	1.000	0.386	0.000	0.000
all	IsolationForest	336	0.811	1.000	0.681	0.000	0.000
all	PCA recon.	1680	0.966	0.961	0.971	0.109	0.989
all	PCA recon.	448	0.000	0.000	0.000	0.109	0.000
all	PCA recon.	1232	0.985	1.000	0.971	0.000	0.000
all	PCA recon.	336	1.000	1.000	1.000	0.000	0.000
all	Conformal residual	1680	0.842	0.984	0.735	0.034	0.947
all	Conformal residual	448	0.000	0.000	0.000	0.034	0.000
all	Conformal residual	1232	0.848	1.000	0.735	0.000	0.000
all	Conformal residual	336	0.800	1.000	0.667	0.000	0.000
all	Calibrated stack	1680	0.986	0.972	1.000	0.080	1.000
all	Calibrated stack	448	0.000	0.000	0.000	0.080	0.000
all	Calibrated stack	1232	1.000	1.000	1.000	0.000	0.000
all	Calibrated stack	336	1.000	1.000	1.000	0.000	0.000
all	Old audit	1680	0.479	0.907	0.326	0.091	0.471
all	Old audit	448	0.000	0.000	0.000	0.091	0.000
all	Old audit	1232	0.491	1.000	0.326	0.000	0.000
all	Old audit	336	0.693	1.000	0.530	0.000	0.000
all	Audit v5	1680	0.477	0.907	0.324	0.091	0.475
all	Audit v5	448	0.000	0.000	0.000	0.091	0.000
all	Audit v5	1232	0.489	1.000	0.324	0.000	0.000
all	Audit v5	336	0.693	1.000	0.530	0.000	0.000
all	Oracle labels	1680	1.000	1.000	1.000	0.000	1.000
all	Oracle labels	448	0.000	0.000	0.000	0.000	0.000
all	Oracle labels	1232	1.000	1.000	1.000	0.000	0.000
all	Oracle labels	336	1.000	1.000	1.000	0.000	0.000

C FULL ABLATION METRICS

Table 10: Full ablation metrics.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
adversarial_compensated_violation	Full v5	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No actuator	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No causality	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No conformal agg.	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No contact	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No energy	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No friction	80	0.797	1.000	0.662	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
adversarial_compensated_violation	No rare-valid	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No support	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	No timestamp	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	Old v4	80	0.740	1.000	0.588	0.000	0.000
adversarial_compensated_violation	Scalar only	80	0.161	1.000	0.087	0.000	0.000
combined_violation_shift	Full v5	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No actuator	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No causality	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No conformal agg.	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No contact	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No energy	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No friction	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No rare-valid	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No support	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	No timestamp	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Old v4	80	1.000	1.000	1.000	0.000	0.000
combined_violation_shift	Scalar only	80	1.000	1.000	1.000	0.000	0.000
mixed_near_threshold	Full v5	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No actuator	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No causality	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No conformal agg.	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No contact	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No energy	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No friction	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No rare-valid	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No support	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	No timestamp	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Old v4	80	0.000	0.000	0.000	0.000	0.000
mixed_near_threshold	Scalar only	80	0.000	0.000	0.000	0.000	0.000
rare_valid_bounce	Full v5	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No actuator	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No causality	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No conformal agg.	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No contact	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No energy	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No friction	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No rare-valid	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No support	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	No timestamp	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	Old v4	80	0.000	0.000	0.000	0.050	0.000
rare_valid_bounce	Scalar only	80	0.000	0.000	0.000	0.050	0.000

D FULL STRESS METRICS

Table 11: Full stress-sweep metrics.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
stress.0.00_adversarial_compensated_violation	Kin. residual	64	0.476	1.000	0.312	0.000	0.000
stress.0.00_adversarial_compensated_violation	Energy residual	64	0.562	1.000	0.391	0.000	0.000
stress.0.00_adversarial_compensated_violation	Max residual	64	0.000	0.000	0.000	0.000	0.000
stress.0.00_adversarial_compensated_violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_adversarial_compensated_violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_adversarial_compensated_violation	PCA recon.	64	0.897	1.000	0.812	0.000	0.000
stress.0.00_adversarial_compensated_violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_adversarial_compensated_violation	Audit v5	64	0.000	0.000	0.000	0.000	0.000
stress.0.00_combined_violation_shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_combined_violation_shift	Energy residual	64	0.000	0.000	0.000	0.000	0.000
stress.0.00_combined_violation_shift	Max residual	64	0.400	1.000	0.250	0.000	0.000
stress.0.00_combined_violation_shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_combined_violation_shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_combined_violation_shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_combined_violation_shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.00_combined_violation_shift	Audit v5	64	0.420	1.000	0.266	0.000	0.000
stress.0.00_rare_valid_bounce	Kin. residual	64	0.000	0.000	0.000	0.328	0.000
stress.0.00_rare_valid_bounce	Energy residual	64	0.000	0.000	0.000	0.359	0.000
stress.0.00_rare_valid_bounce	Max residual	64	0.000	0.000	0.000	0.000	0.000
stress.0.00_rare_valid_bounce	HGB classifier	64	0.000	0.000	0.000	0.328	0.000
stress.0.00_rare_valid_bounce	RF ensemble	64	0.000	0.000	0.000	0.297	0.000
stress.0.00_rare_valid_bounce	PCA recon.	64	0.000	0.000	0.000	0.500	0.000
stress.0.00_rare_valid_bounce	Calibrated stack	64	0.000	0.000	0.000	0.375	0.000
stress.0.00_rare_valid_bounce	Audit v5	64	0.000	0.000	0.000	0.000	0.000
stress.0.20_adversarial_compensated_violation	Kin. residual	64	0.400	1.000	0.250	0.000	0.000
stress.0.20_adversarial_compensated_violation	Energy residual	64	0.458	1.000	0.297	0.000	0.000
stress.0.20_adversarial_compensated_violation	Max residual	64	0.247	1.000	0.141	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
stress.0.20.adversarial.compensated.violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.adversarial.compensated.violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.adversarial.compensated.violation	PCA recon.	64	0.984	1.000	0.969	0.000	0.000
stress.0.20.adversarial.compensated.violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.adversarial.compensated.violation	Audit v5	64	0.247	1.000	0.141	0.000	0.000
stress.0.20.combined.violation.shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.combined.violation.shift	Energy residual	64	0.000	0.000	0.000	0.000	0.000
stress.0.20.combined.violation.shift	Max residual	64	0.836	1.000	0.719	0.000	0.000
stress.0.20.combined.violation.shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.combined.violation.shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.combined.violation.shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.combined.violation.shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.20.combined.violation.shift	Audit v5	64	0.815	1.000	0.688	0.000	0.000
stress.0.20.rare.valid.bounce	Kin. residual	64	0.000	0.000	0.000	0.250	0.000
stress.0.20.rare.valid.bounce	Energy residual	64	0.000	0.000	0.000	0.297	0.000
stress.0.20.rare.valid.bounce	Max residual	64	0.000	0.000	0.000	0.000	0.000
stress.0.20.rare.valid.bounce	HGB classifier	64	0.000	0.000	0.000	0.219	0.000
stress.0.20.rare.valid.bounce	RF ensemble	64	0.000	0.000	0.000	0.172	0.000
stress.0.20.rare.valid.bounce	PCA recon.	64	0.000	0.000	0.000	0.391	0.000
stress.0.20.rare.valid.bounce	Calibrated stack	64	0.000	0.000	0.000	0.266	0.000
stress.0.20.rare.valid.bounce	Audit v5	64	0.000	0.000	0.000	0.000	0.000
stress.0.40.adversarial.compensated.violation	Kin. residual	64	0.757	1.000	0.609	0.000	0.000
stress.0.40.adversarial.compensated.violation	Energy residual	64	0.529	1.000	0.359	0.000	0.000
stress.0.40.adversarial.compensated.violation	Max residual	64	0.476	1.000	0.312	0.000	0.000
stress.0.40.adversarial.compensated.violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.adversarial.compensated.violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.adversarial.compensated.violation	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.adversarial.compensated.violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.adversarial.compensated.violation	Audit v5	64	0.476	1.000	0.312	0.000	0.000
stress.0.40.combined.violation.shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	Energy residual	64	0.529	1.000	0.359	0.000	0.000
stress.0.40.combined.violation.shift	Max residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.combined.violation.shift	Audit v5	64	1.000	1.000	1.000	0.000	0.000
stress.0.40.rare.valid.bounce	Kin. residual	64	0.000	0.000	0.000	0.250	0.000
stress.0.40.rare.valid.bounce	Energy residual	64	0.000	0.000	0.000	0.328	0.000
stress.0.40.rare.valid.bounce	Max residual	64	0.000	0.000	0.000	0.016	0.000
stress.0.40.rare.valid.bounce	HGB classifier	64	0.000	0.000	0.000	0.266	0.000
stress.0.40.rare.valid.bounce	RF ensemble	64	0.000	0.000	0.000	0.141	0.000
stress.0.40.rare.valid.bounce	PCA recon.	64	0.000	0.000	0.000	0.578	0.000
stress.0.40.rare.valid.bounce	Calibrated stack	64	0.000	0.000	0.000	0.297	0.000
stress.0.40.rare.valid.bounce	Audit v5	64	0.000	0.000	0.000	0.031	0.000
stress.0.60.adversarial.compensated.violation	Kin. residual	64	0.968	1.000	0.938	0.000	0.000
stress.0.60.adversarial.compensated.violation	Energy residual	64	0.512	1.000	0.344	0.000	0.000
stress.0.60.adversarial.compensated.violation	Max residual	64	0.609	1.000	0.438	0.000	0.000
stress.0.60.adversarial.compensated.violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.adversarial.compensated.violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.adversarial.compensated.violation	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.adversarial.compensated.violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.adversarial.compensated.violation	Audit v5	64	0.609	1.000	0.438	0.000	0.000
stress.0.60.combined.violation.shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	Energy residual	64	0.968	1.000	0.938	0.000	0.000
stress.0.60.combined.violation.shift	Max residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.combined.violation.shift	Audit v5	64	1.000	1.000	1.000	0.000	0.000
stress.0.60.rare.valid.bounce	Kin. residual	64	0.000	0.000	0.000	0.391	0.000
stress.0.60.rare.valid.bounce	Energy residual	64	0.000	0.000	0.000	0.297	0.000
stress.0.60.rare.valid.bounce	Max residual	64	0.000	0.000	0.000	0.109	0.000
stress.0.60.rare.valid.bounce	HGB classifier	64	0.000	0.000	0.000	0.297	0.000
stress.0.60.rare.valid.bounce	RF ensemble	64	0.000	0.000	0.000	0.203	0.000
stress.0.60.rare.valid.bounce	PCA recon.	64	0.000	0.000	0.000	0.562	0.000
stress.0.60.rare.valid.bounce	Calibrated stack	64	0.000	0.000	0.000	0.281	0.000
stress.0.60.rare.valid.bounce	Audit v5	64	0.000	0.000	0.000	0.109	0.000
stress.0.80.adversarial.compensated.violation	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.adversarial.compensated.violation	Energy residual	64	0.529	1.000	0.359	0.000	0.000
stress.0.80.adversarial.compensated.violation	Max residual	64	0.707	1.000	0.547	0.000	0.000
stress.0.80.adversarial.compensated.violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.adversarial.compensated.violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.adversarial.compensated.violation	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.adversarial.compensated.violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.adversarial.compensated.violation	Audit v5	64	0.707	1.000	0.547	0.000	0.000
stress.0.80.combined.violation.shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress.0.80.combined.violation.shift	Energy residual	64	1.000	1.000	1.000	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
stress_0.80_combined_violation_shift	Max residual	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_combined_violation_shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_combined_violation_shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_combined_violation_shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_combined_violation_shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_combined_violation_shift	Audit v5	64	1.000	1.000	1.000	0.000	0.000
stress_0.80_rare_valid_bounce	Kin. residual	64	0.000	0.000	0.000	0.484	0.000
stress_0.80_rare_valid_bounce	Energy residual	64	0.000	0.000	0.000	0.281	0.000
stress_0.80_rare_valid_bounce	Max residual	64	0.000	0.000	0.000	0.156	0.000
stress_0.80_rare_valid_bounce	HGB classifier	64	0.000	0.000	0.000	0.500	0.000
stress_0.80_rare_valid_bounce	RF ensemble	64	0.000	0.000	0.000	0.484	0.000
stress_0.80_rare_valid_bounce	PCA recon.	64	0.000	0.000	0.000	0.812	0.000
stress_0.80_rare_valid_bounce	Calibrated stack	64	0.000	0.000	0.000	0.531	0.000
stress_0.80_rare_valid_bounce	Audit v5	64	0.000	0.000	0.000	0.156	0.000
stress_1.00_adversarial_compensated_violation	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_adversarial_compensated_violation	Energy residual	64	0.609	1.000	0.438	0.000	0.000
stress_1.00_adversarial_compensated_violation	Max residual	64	0.847	1.000	0.734	0.000	0.000
stress_1.00_adversarial_compensated_violation	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_adversarial_compensated_violation	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_adversarial_compensated_violation	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_adversarial_compensated_violation	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_adversarial_compensated_violation	Audit v5	64	0.847	1.000	0.734	0.000	0.000
stress_1.00_combined_violation_shift	Kin. residual	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	Energy residual	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	Max residual	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	HGB classifier	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	RF ensemble	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	PCA recon.	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	Calibrated stack	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_combined_violation_shift	Audit v5	64	1.000	1.000	1.000	0.000	0.000
stress_1.00_rare_valid_bounce	Kin. residual	64	0.000	0.000	0.000	0.391	0.000
stress_1.00_rare_valid_bounce	Energy residual	64	0.000	0.000	0.000	0.266	0.000
stress_1.00_rare_valid_bounce	Max residual	64	0.000	0.000	0.000	0.266	0.000
stress_1.00_rare_valid_bounce	HGB classifier	64	0.000	0.000	0.000	0.453	0.000
stress_1.00_rare_valid_bounce	RF ensemble	64	0.000	0.000	0.000	0.328	0.000
stress_1.00_rare_valid_bounce	PCA recon.	64	0.000	0.000	0.000	0.875	0.000
stress_1.00_rare_valid_bounce	Calibrated stack	64	0.000	0.000	0.000	0.422	0.000
stress_1.00_rare_valid_bounce	Audit v5	64	0.000	0.000	0.000	0.266	0.000

E PAIRWISE SEED COMPARISONS

Table 12: All split-level paired seed comparisons versus audit v5. Positive values favor audit v5.

Split	Baseline	Diff	t approx.	p approx.
nominal_valid	Random	0.000	0.000	1.000
nominal_valid	Kin. residual	0.000	0.000	1.000
nominal_valid	Energy residual	0.000	0.000	1.000
nominal_valid	Contact residual	0.000	0.000	1.000
nominal_valid	Max residual	0.000	0.000	1.000
nominal_valid	Logistic stack	0.000	0.000	1.000
nominal_valid	HGB classifier	0.000	0.000	1.000
nominal_valid	RF ensemble	0.000	0.000	1.000
nominal_valid	IsolationForest	0.000	0.000	1.000
nominal_valid	PCA recon.	0.000	0.000	1.000
nominal_valid	Conformal residual	0.000	0.000	1.000
nominal_valid	Calibrated stack	0.000	0.000	1.000
nominal_valid	Old audit	0.000	0.000	1.000
nominal_valid	Oracle labels	0.000	0.000	1.000
rare_valid_bounce	Random	0.000	0.000	1.000
rare_valid_bounce	Kin. residual	0.000	0.000	1.000
rare_valid_bounce	Energy residual	0.000	0.000	1.000
rare_valid_bounce	Contact residual	0.000	0.000	1.000
rare_valid_bounce	Max residual	0.000	0.000	1.000
rare_valid_bounce	Logistic stack	0.000	0.000	1.000
rare_valid_bounce	HGB classifier	0.000	0.000	1.000
rare_valid_bounce	RF ensemble	0.000	0.000	1.000
rare_valid_bounce	IsolationForest	0.000	0.000	1.000
rare_valid_bounce	PCA recon.	0.000	0.000	1.000
rare_valid_bounce	Conformal residual	0.000	0.000	1.000
rare_valid_bounce	Calibrated stack	0.000	0.000	1.000
rare_valid_bounce	Old audit	0.000	0.000	1.000
rare_valid_bounce	Oracle labels	0.000	0.000	1.000
valid_clock_skew	Random	0.000	0.000	1.000
valid_clock_skew	Kin. residual	0.000	0.000	1.000

	Split	Baseline	Diff	t approx.	p approx.
756					
757					
758	valid_clock_skew	Energy residual	0.000	0.000	1.000
759	valid_clock_skew	Contact residual	0.000	0.000	1.000
760	valid_clock_skew	Max residual	0.000	0.000	1.000
761	valid_clock_skew	Logistic stack	0.000	0.000	1.000
762	valid_clock_skew	HGB classifier	0.000	0.000	1.000
763	valid_clock_skew	RF ensemble	0.000	0.000	1.000
764	valid_clock_skew	IsolationForest	0.000	0.000	1.000
765	valid_clock_skew	PCA recon.	0.000	0.000	1.000
766	valid_clock_skew	Conformal residual	0.000	0.000	1.000
767	valid_clock_skew	Calibrated stack	0.000	0.000	1.000
768	valid_clock_skew	Old audit	0.000	0.000	1.000
769	valid_clock_skew	Oracle labels	0.000	0.000	1.000
770	valid_low_friction_slip	Random	0.000	0.000	1.000
771	valid_low_friction_slip	Kin. residual	0.000	0.000	1.000
772	valid_low_friction_slip	Energy residual	0.000	0.000	1.000
773	valid_low_friction_slip	Contact residual	0.000	0.000	1.000
774	valid_low_friction_slip	Max residual	0.000	0.000	1.000
775	valid_low_friction_slip	Logistic stack	0.000	0.000	1.000
776	valid_low_friction_slip	HGB classifier	0.000	0.000	1.000
777	valid_low_friction_slip	RF ensemble	0.000	0.000	1.000
778	valid_low_friction_slip	IsolationForest	0.000	0.000	1.000
779	valid_low_friction_slip	PCA recon.	0.000	0.000	1.000
780	valid_low_friction_slip	Conformal residual	0.000	0.000	1.000
781	valid_low_friction_slip	Calibrated stack	0.000	0.000	1.000
782	valid_low_friction_slip	Old audit	0.000	0.000	1.000
783	valid_low_friction_slip	Oracle labels	0.000	0.000	1.000
784	contact_corruption	Random	-0.755	-21.478	0.000
785	contact_corruption	Kin. residual	0.000	0.000	1.000
786	contact_corruption	Energy residual	0.000	0.000	1.000
787	contact_corruption	Contact residual	-1.000	-1000000.000	0.000
788	contact_corruption	Max residual	0.000	0.000	1.000
789	contact_corruption	Logistic stack	-1.000	-1000000.000	0.000
790	contact_corruption	HGB classifier	-1.000	-1000000.000	0.000
791	contact_corruption	RF ensemble	-1.000	-1000000.000	0.000
792	contact_corruption	IsolationForest	-1.000	-1000000.000	0.000
793	contact_corruption	PCA recon.	-1.000	-1000000.000	0.000
794	contact_corruption	Conformal residual	-1.000	-1000000.000	0.000
795	contact_corruption	Calibrated stack	-1.000	-1000000.000	0.000
796	contact_corruption	Old audit	0.000	0.000	1.000
797	contact_corruption	Oracle labels	-1.000	-1000000.000	0.000
798	energy_work_corruption	Random	-0.351	-5.245	0.000
799	energy_work_corruption	Kin. residual	-0.655	-9.519	0.000
800	energy_work_corruption	Energy residual	-0.655	-9.519	0.000
801	energy_work_corruption	Contact residual	0.345	5.009	0.000
802	energy_work_corruption	Max residual	0.000	0.000	1.000
803	energy_work_corruption	Logistic stack	-0.591	-6.806	0.000
804	energy_work_corruption	HGB classifier	-0.655	-9.519	0.000
805	energy_work_corruption	RF ensemble	-0.655	-9.519	0.000
806	energy_work_corruption	IsolationForest	0.045	0.413	0.679
807	energy_work_corruption	PCA recon.	-0.655	-9.519	0.000
808	energy_work_corruption	Conformal residual	-0.655	-9.519	0.000
809	energy_work_corruption	Calibrated stack	-0.655	-9.519	0.000
810	energy_work_corruption	Old audit	0.000	0.000	1.000
811	energy_work_corruption	Oracle labels	-0.655	-9.519	0.000
812	support_levitation	Random	-0.674	-11.845	0.000
813	support_levitation	Kin. residual	-1.000	-1000000.000	0.000
814	support_levitation	Energy residual	-0.490	-8.237	0.000
815	support_levitation	Contact residual	0.000	0.000	1.000
816	support_levitation	Max residual	0.000	0.000	1.000
817	support_levitation	Logistic stack	-1.000	-1000000.000	0.000
818	support_levitation	HGB classifier	-1.000	-1000000.000	0.000
819	support_levitation	RF ensemble	-1.000	-1000000.000	0.000
820	support_levitation	IsolationForest	-0.017	-1.000	0.317
821	support_levitation	PCA recon.	-1.000	-1000000.000	0.000
822	support_levitation	Conformal residual	-1.000	-1000000.000	0.000
823	support_levitation	Calibrated stack	-1.000	-1000000.000	0.000
824	support_levitation	Old audit	0.000	0.000	1.000
825	support_levitation	Oracle labels	-1.000	-1000000.000	0.000
826	actuator_saturation	Random	-0.341	-4.778	0.000
827	actuator_saturation	Kin. residual	0.345	4.957	0.000
828	actuator_saturation	Energy residual	0.345	4.957	0.000
829	actuator_saturation	Contact residual	0.329	4.604	0.000
830	actuator_saturation	Max residual	-0.017	-1.000	0.317
831	actuator_saturation	Logistic stack	-0.528	-5.278	0.000
832	actuator_saturation	HGB classifier	-0.655	-9.397	0.000
833	actuator_saturation	RF ensemble	-0.655	-9.397	0.000
834	actuator_saturation	IsolationForest	0.345	4.957	0.000
835	actuator_saturation	PCA recon.	-0.655	-9.397	0.000
836	actuator_saturation	Conformal residual	-0.429	-4.839	0.000

Split	Baseline	Diff	t approx.	p approx.
actuator_saturation	Calibrated stack	-0.655	-9.397	0.000
actuator_saturation	Old audit	-0.017	-1.000	0.317
actuator_saturation	Oracle labels	-0.655	-9.397	0.000
noncausal_teleport	Random	0.336	7.834	0.000
noncausal_teleport	Kin. residual	0.000	0.000	1.000
noncausal_teleport	Energy residual	1.000	1000000.000	0.000
noncausal_teleport	Contact residual	1.000	1000000.000	0.000
noncausal_teleport	Max residual	0.000	0.000	1.000
noncausal_teleport	Logistic stack	0.005	1.000	0.317
noncausal_teleport	HGB classifier	0.000	0.000	1.000
noncausal_teleport	RF ensemble	0.000	0.000	1.000
noncausal_teleport	IsolationForest	1.000	1000000.000	0.000
noncausal_teleport	PCA recon.	0.000	0.000	1.000
noncausal_teleport	Conformal residual	0.000	0.000	1.000
noncausal_teleport	Calibrated stack	0.000	0.000	1.000
noncausal_teleport	Old audit	0.000	0.000	1.000
noncausal_teleport	Oracle labels	0.000	0.000	1.000
combined_violation_shift	Random	0.368	16.792	0.000
combined_violation_shift	Kin. residual	0.000	0.000	1.000
combined_violation_shift	Energy residual	0.000	0.000	1.000
combined_violation_shift	Contact residual	1.000	1000000.000	0.000
combined_violation_shift	Max residual	0.000	0.000	1.000
combined_violation_shift	Logistic stack	0.000	0.000	1.000
combined_violation_shift	HGB classifier	0.000	0.000	1.000
combined_violation_shift	RF ensemble	0.000	0.000	1.000
combined_violation_shift	IsolationForest	0.000	0.000	1.000
combined_violation_shift	PCA recon.	0.000	0.000	1.000
combined_violation_shift	Conformal residual	0.000	0.000	1.000
combined_violation_shift	Calibrated stack	0.000	0.000	1.000
combined_violation_shift	Old audit	0.000	0.000	1.000
combined_violation_shift	Oracle labels	0.000	0.000	1.000
subtle_contact_corruption	Random	-0.725	-16.581	0.000
subtle_contact_corruption	Kin. residual	0.000	0.000	1.000
subtle_contact_corruption	Energy residual	0.000	0.000	1.000
subtle_contact_corruption	Contact residual	-1.000	-1000000.000	0.000
subtle_contact_corruption	Max residual	0.000	0.000	1.000
subtle_contact_corruption	Logistic stack	-0.951	-50.313	0.000
subtle_contact_corruption	HGB classifier	-1.000	-1000000.000	0.000
subtle_contact_corruption	RF ensemble	-1.000	-1000000.000	0.000
subtle_contact_corruption	IsolationForest	-1.000	-1000000.000	0.000
subtle_contact_corruption	PCA recon.	-1.000	-1000000.000	0.000
subtle_contact_corruption	Conformal residual	-1.000	-1000000.000	0.000
subtle_contact_corruption	Calibrated stack	-1.000	-1000000.000	0.000
subtle_contact_corruption	Old audit	0.000	0.000	1.000
subtle_contact_corruption	Oracle labels	-1.000	-1000000.000	0.000
subtle_energy_corruption	Random	-0.720	-13.470	0.000
subtle_energy_corruption	Kin. residual	0.000	0.000	1.000
subtle_energy_corruption	Energy residual	0.000	0.000	1.000
subtle_energy_corruption	Contact residual	-0.033	-1.528	0.127
subtle_energy_corruption	Max residual	0.000	0.000	1.000
subtle_energy_corruption	Logistic stack	-0.017	-1.000	0.317
subtle_energy_corruption	HGB classifier	-1.000	-1000000.000	0.000
subtle_energy_corruption	RF ensemble	-1.000	-1000000.000	0.000
subtle_energy_corruption	IsolationForest	0.000	0.000	1.000
subtle_energy_corruption	PCA recon.	-0.806	-37.451	0.000
subtle_energy_corruption	Conformal residual	0.000	0.000	1.000
subtle_energy_corruption	Calibrated stack	-1.000	-1000000.000	0.000
subtle_energy_corruption	Old audit	0.000	0.000	1.000
subtle_energy_corruption	Oracle labels	-1.000	-1000000.000	0.000
timestamp_noncausal_corruption	Random	-0.017	-0.302	0.763
timestamp_noncausal_corruption	Kin. residual	0.510	13.448	0.000
timestamp_noncausal_corruption	Energy residual	0.681	16.056	0.000
timestamp_noncausal_corruption	Contact residual	0.681	16.056	0.000
timestamp_noncausal_corruption	Max residual	-0.006	-1.000	0.317
timestamp_noncausal_corruption	Logistic stack	-0.128	-1.970	0.049
timestamp_noncausal_corruption	HGB classifier	-0.319	-7.520	0.000
timestamp_noncausal_corruption	RF ensemble	-0.319	-7.520	0.000
timestamp_noncausal_corruption	IsolationForest	0.681	16.056	0.000
timestamp_noncausal_corruption	PCA recon.	-0.319	-7.520	0.000
timestamp_noncausal_corruption	Conformal residual	0.072	1.175	0.240
timestamp_noncausal_corruption	Calibrated stack	-0.319	-7.520	0.000
timestamp_noncausal_corruption	Old audit	-0.006	-1.000	0.317
timestamp_noncausal_corruption	Oracle labels	-0.319	-7.520	0.000
adversarial_compensated_violation	Random	0.117	1.554	0.120
adversarial_compensated_violation	Kin. residual	-0.271	-5.622	0.000
adversarial_compensated_violation	Energy residual	0.260	2.568	0.010
adversarial_compensated_violation	Contact residual	-0.271	-5.622	0.000
adversarial_compensated_violation	Max residual	0.000	0.000	1.000
adversarial_compensated_violation	Logistic stack	-0.261	-5.098	0.000

Split	Baseline	Diff	t approx.	p approx.
adversarial_compensated_violation	HGB classifier	-0.271	-5.622	0.000
adversarial_compensated_violation	RF ensemble	-0.271	-5.622	0.000
adversarial_compensated_violation	IsolationForest	-0.271	-5.622	0.000
adversarial_compensated_violation	PCA recon.	-0.271	-5.622	0.000
adversarial_compensated_violation	Conformal residual	-0.271	-5.622	0.000
adversarial_compensated_violation	Calibrated stack	-0.271	-5.622	0.000
adversarial_compensated_violation	Old audit	0.000	0.000	1.000
adversarial_compensated_violation	Oracle labels	-0.271	-5.622	0.000
mixed_near_threshold	Random	-0.637	-11.725	0.000
mixed_near_threshold	Kin. residual	-0.952	-57.270	0.000
mixed_near_threshold	Energy residual	-0.017	-1.000	0.317
mixed_near_threshold	Contact residual	0.000	0.000	1.000
mixed_near_threshold	Max residual	0.000	0.000	1.000
mixed_near_threshold	Logistic stack	-1.000	-1000000.000	0.000
mixed_near_threshold	HGB classifier	-1.000	-1000000.000	0.000
mixed_near_threshold	RF ensemble	-1.000	-1000000.000	0.000
mixed_near_threshold	IsolationForest	-0.079	-1.946	0.052
mixed_near_threshold	PCA recon.	-1.000	-1000000.000	0.000
mixed_near_threshold	Conformal residual	0.000	0.000	1.000
mixed_near_threshold	Calibrated stack	-1.000	-1000000.000	0.000
mixed_near_threshold	Old audit	0.000	0.000	1.000
mixed_near_threshold	Oracle labels	-1.000	-1000000.000	0.000

F SELECTED SEED METRICS

Table 13: Selected seed-level metrics for reproducibility.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
actuator_saturation	Audit v5	14	0.600	1.000	0.429	0.000	0.000
actuator_saturation	Audit v5	14	0.133	1.000	0.071	0.000	0.000
actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000
actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000
actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000
actuator_saturation	Audit v5	14	0.526	1.000	0.357	0.000	0.000
actuator_saturation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Audit v5	14	0.444	1.000	0.286	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
adversarial_compensated_violation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000

[illegible]

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
972								
973	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
974	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
975	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
976	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
977	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
978	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
979	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
980	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
981	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
982	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
983	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
984	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
985	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
986	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
987	energy_work_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
988	energy_work_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
989	energy_work_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
990	energy_work_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
991	energy_work_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
992	energy_work_corruption	Audit v5	14	0.526	1.000	0.357	0.000	0.000
993	energy_work_corruption	Audit v5	14	0.353	1.000	0.214	0.000	0.000
994	energy_work_corruption	Audit v5	14	0.133	1.000	0.071	0.000	0.000
995	energy_work_corruption	Audit v5	14	0.600	1.000	0.429	0.000	0.000
996	energy_work_corruption	Audit v5	14	0.526	1.000	0.357	0.000	0.000
997	energy_work_corruption	Audit v5	14	0.133	1.000	0.071	0.000	0.000
998	energy_work_corruption	Audit v5	14	0.353	1.000	0.214	0.000	0.000
999	mixed_near_threshold	Kin. residual	14	0.880	1.000	0.786	0.000	0.000
1000	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1001	mixed_near_threshold	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1002	mixed_near_threshold	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1003	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1004	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1005	mixed_near_threshold	Kin. residual	14	0.880	1.000	0.786	0.000	0.000
1006	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1007	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1008	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1009	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1010	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1011	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1012	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1013	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1014	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1015	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1016	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1017	mixed_near_threshold	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1018	mixed_near_threshold	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1026								
1027								
1028	nominal_valid	HGB classifier	14	0.000	0.000	0.000	0.071	0.000
1028	nominal_valid	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
1029	nominal_valid	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
1029	nominal_valid	HGB classifier	14	0.000	0.000	0.000	0.071	0.000
1030	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.214	0.000
1031	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
1031	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
1032	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
1033	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
1033	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
1034	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
1035	nominal_valid	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
1036	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1036	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1037	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1037	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1038	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1039	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1039	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1040	nominal_valid	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1041	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1042	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1042	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1043	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1044	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1044	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1045	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1046	noncausal_teleport	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1047	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1048	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1049	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1049	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1050	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1051	noncausal_teleport	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1052	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1053	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1054	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1055	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1056	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1057	noncausal_teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1058	noncausal_teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1059	noncausal_teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1060	noncausal_teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1061	noncausal_teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1062	noncausal_teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1063	rare_valid_bounce	Kin. residual	14	0.000	0.000	0.000	0.357	0.000
1064	rare_valid_bounce	Kin. residual	14	0.000	0.000	0.000	0.143	0.000
1065	rare_valid_bounce	Kin. residual	14	0.000	0.000	0.000	0.214	0.000
1066	rare_valid_bounce	Kin. residual	14	0.000	0.000	0.000	0.214	0.000
1067	rare_valid_bounce	Kin. residual	14	0.000	0.000	0.000	0.286	0.000
1068	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.357	0.000
1069	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
1070	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
1071	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.214	0.000
1072	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
1073	rare_valid_bounce	HGB classifier	14	0.000	0.000	0.000	0.286	0.000
1074	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.429	0.000
1075	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
1076	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.286	0.000
1077	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.143	0.000
1078	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.214	0.000
1079	rare_valid_bounce	Calibrated stack	14	0.000	0.000	0.000	0.286	0.000
1079	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1079	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1079	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.143	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1080								
1081	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1082	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.286	0.000
1083	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.071	0.000
1084	rare_valid_bounce	Audit v5	14	0.000	0.000	0.000	0.071	0.000
1085	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1086	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1087	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1088	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1089	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1090	subtle_contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1091	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1092	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1093	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1094	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1095	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1096	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1097	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1098	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1099	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1100	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1101	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1102	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1103	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1104	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1105	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1106	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1107	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1108	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1109	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1110	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1111	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1112	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1113	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1114	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1115	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1116	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1117	subtle_energy_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1118	subtle_energy_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1119	subtle_energy_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1120	subtle_energy_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1121	subtle_energy_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1122	subtle_energy_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1123	subtle_energy_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1124	subtle_energy_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1125	subtle_energy_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1126	subtle_energy_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1127	subtle_energy_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1128	support_levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1129	support_levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1130	support_levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1131	support_levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1132	support_levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1133	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1134								
1135	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1136	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1137	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1138	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1139	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1140	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1141	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1142	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1143	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1144	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1145	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1146	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1147	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1148	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1149	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1150	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1151	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
1152	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
1153	timestamp_noncausal_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1154	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
1155	timestamp_noncausal_corruption	Kin. residual	14	0.133	1.000	0.071	0.000	0.000
1156	timestamp_noncausal_corruption	Kin. residual	14	0.133	1.000	0.071	0.000	0.000
1157	timestamp_noncausal_corruption	Kin. residual	14	0.353	1.000	0.214	0.000	0.000
1158	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1159	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1160	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1161	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1162	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1163	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1164	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1165	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1166	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1167	timestamp_noncausal_corruption	Audit v5	14	0.667	1.000	0.500	0.000	0.000
1168	timestamp_noncausal_corruption	Audit v5	14	0.667	1.000	0.500	0.000	0.000
1169	timestamp_noncausal_corruption	Audit v5	14	0.444	1.000	0.286	0.000	0.000
1170	timestamp_noncausal_corruption	Audit v5	14	0.600	1.000	0.429	0.000	0.000
1171	timestamp_noncausal_corruption	Audit v5	14	0.727	1.000	0.571	0.000	0.000
1172	timestamp_noncausal_corruption	Audit v5	14	0.833	1.000	0.714	0.000	0.000
1173	timestamp_noncausal_corruption	Audit v5	14	0.727	1.000	0.571	0.000	0.000
1174	timestamp_noncausal_corruption	Audit v5	14	0.783	1.000	0.643	0.000	0.000
1175	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1176	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1177	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1178	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1179	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1180	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1181	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1182	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1183	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1184	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1185	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1186	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1187	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.214	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.071	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.214	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.143	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.214	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_clock_skew	Audit v5	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.357	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.500	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.429	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.286	0.000

G ALL SEED METRICS

Table 14: All method/split/seed metrics from the frozen run.

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
actuator_saturation	Random	14	0.600	1.000	0.429	0.000	0.000
actuator_saturation	Random	14	0.727	1.000	0.571	0.000	0.000
actuator_saturation	Random	14	0.667	1.000	0.500	0.000	0.000
actuator_saturation	Random	14	0.783	1.000	0.643	0.000	0.000
actuator_saturation	Random	14	0.600	1.000	0.429	0.000	0.000
actuator_saturation	Random	14	0.727	1.000	0.571	0.000	0.000
actuator_saturation	Random	14	0.600	1.000	0.429	0.000	0.000
actuator_saturation	Random	14	0.783	1.000	0.643	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Energy residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Contact residual	14	0.133	1.000	0.071	0.000	0.000
actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1242								
1243								
1244	actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
	actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1245	actuator_saturation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1246	actuator_saturation	Max residual	14	0.600	1.000	0.429	0.000	0.000
	actuator_saturation	Max residual	14	0.133	1.000	0.071	0.000	0.000
1247	actuator_saturation	Max residual	14	0.353	1.000	0.214	0.000	0.000
	actuator_saturation	Max residual	14	0.353	1.000	0.214	0.000	0.000
1248	actuator_saturation	Max residual	14	0.353	1.000	0.214	0.000	0.000
	actuator_saturation	Max residual	14	0.526	1.000	0.357	0.000	0.000
1249	actuator_saturation	Max residual	14	0.133	1.000	0.071	0.000	0.000
	actuator_saturation	Max residual	14	0.444	1.000	0.286	0.000	0.000
1250	actuator_saturation	Logistic stack	14	0.783	1.000	0.643	0.000	0.000
1251	actuator_saturation	Logistic stack	14	0.963	1.000	0.929	0.000	0.000
1252	actuator_saturation	Logistic stack	14	0.923	1.000	0.857	0.000	0.000
1253	actuator_saturation	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
	actuator_saturation	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
1254	actuator_saturation	Logistic stack	14	0.727	1.000	0.571	0.000	0.000
1255	actuator_saturation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	Logistic stack	14	0.833	1.000	0.714	0.000	0.000
1256	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1257	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1258	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1259	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1260	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1261	actuator_saturation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1262	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1263	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1264	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1265	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1266	actuator_saturation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1267	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1268	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1269	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1270	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1271	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
	actuator_saturation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1272	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1273	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1274	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1275	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1276	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1277	actuator_saturation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	Conformal residual	14	0.880	1.000	0.786	0.000	0.000
1278	actuator_saturation	Conformal residual	14	0.833	1.000	0.714	0.000	0.000
1279	actuator_saturation	Conformal residual	14	0.833	1.000	0.714	0.000	0.000
	actuator_saturation	Conformal residual	14	0.783	1.000	0.643	0.000	0.000
1280	actuator_saturation	Conformal residual	14	0.600	1.000	0.429	0.000	0.000
1281	actuator_saturation	Conformal residual	14	0.783	1.000	0.643	0.000	0.000
	actuator_saturation	Conformal residual	14	0.880	1.000	0.786	0.000	0.000
1282	actuator_saturation	Conformal residual	14	0.600	1.000	0.429	0.000	0.000
1283	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1284	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1285	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1286	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1287	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
	actuator_saturation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1288	actuator_saturation	Old audit	14	0.600	1.000	0.429	0.000	0.000
1289	actuator_saturation	Old audit	14	0.133	1.000	0.071	0.000	0.000
	actuator_saturation	Old audit	14	0.353	1.000	0.214	0.000	0.000
1290	actuator_saturation	Old audit	14	0.353	1.000	0.214	0.000	0.000
1291	actuator_saturation	Old audit	14	0.353	1.000	0.214	0.000	0.000
	actuator_saturation	Old audit	14	0.526	1.000	0.357	0.000	0.000
1292	actuator_saturation	Old audit	14	0.133	1.000	0.071	0.000	0.000
1293	actuator_saturation	Old audit	14	0.444	1.000	0.286	0.000	0.000
	actuator_saturation	Audit v5	14	0.600	1.000	0.429	0.000	0.000
1294	actuator_saturation	Audit v5	14	0.133	1.000	0.071	0.000	0.000
1295	actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000
	actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1296								
1297	actuator_saturation	Audit v5	14	0.353	1.000	0.214	0.000	0.000
1298	actuator_saturation	Audit v5	14	0.526	1.000	0.357	0.000	0.000
1299	actuator_saturation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1300	actuator_saturation	Audit v5	14	0.444	1.000	0.286	0.000	0.000
1301	actuator_saturation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1302	actuator_saturation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1303	actuator_saturation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1304	actuator_saturation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1305	actuator_saturation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1306	adversarial_compensated_violation	Random	14	0.526	1.000	0.357	0.000	0.000
1307	adversarial_compensated_violation	Random	14	0.667	1.000	0.500	0.000	0.000
1308	adversarial_compensated_violation	Random	14	0.353	1.000	0.214	0.000	0.000
1309	adversarial_compensated_violation	Random	14	0.526	1.000	0.357	0.000	0.000
1310	adversarial_compensated_violation	Random	14	0.600	1.000	0.429	0.000	0.000
1311	adversarial_compensated_violation	Random	14	0.833	1.000	0.714	0.000	0.000
1312	adversarial_compensated_violation	Random	14	0.727	1.000	0.571	0.000	0.000
1313	adversarial_compensated_violation	Random	14	0.667	1.000	0.500	0.000	0.000
1314	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1315	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1316	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1317	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1318	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1319	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1320	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1321	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1322	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1323	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1324	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1325	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1326	adversarial_compensated_violation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1327	adversarial_compensated_violation	Energy residual	14	0.783	1.000	0.643	0.000	0.000
1328	adversarial_compensated_violation	Energy residual	14	0.250	1.000	0.143	0.000	0.000
1329	adversarial_compensated_violation	Energy residual	14	0.353	1.000	0.214	0.000	0.000
1330	adversarial_compensated_violation	Energy residual	14	0.526	1.000	0.357	0.000	0.000
1331	adversarial_compensated_violation	Energy residual	14	0.353	1.000	0.214	0.000	0.000
1332	adversarial_compensated_violation	Energy residual	14	0.600	1.000	0.429	0.000	0.000
1333	adversarial_compensated_violation	Energy residual	14	0.444	1.000	0.286	0.000	0.000
1334	adversarial_compensated_violation	Energy residual	14	0.444	1.000	0.286	0.000	0.000
1335	adversarial_compensated_violation	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1336	adversarial_compensated_violation	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1337	adversarial_compensated_violation	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1338	adversarial_compensated_violation	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1339	adversarial_compensated_violation	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1340	adversarial_compensated_violation	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1341	adversarial_compensated_violation	Contact residual						

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1350								
1351	adversarial.compensated.violation	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1352	adversarial.compensated.violation	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1353	adversarial.compensated.violation	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1354	adversarial.compensated.violation	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1355	adversarial.compensated.violation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1356	adversarial.compensated.violation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1357	adversarial.compensated.violation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1358	adversarial.compensated.violation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1359	adversarial.compensated.violation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1360	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1361	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1362	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1363	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1364	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1365	adversarial.compensated.violation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1366	adversarial.compensated.violation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1367	adversarial.compensated.violation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1368	adversarial.compensated.violation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1369	adversarial.compensated.violation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1370	adversarial.compensated.violation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1371	adversarial.compensated.violation	Old audit	14	0.526	1.000	0.357	0.000	0.000
1372	adversarial.compensated.violation	Old audit	14	0.833	1.000	0.714	0.000	0.000
1373	adversarial.compensated.violation	Old audit	14	0.727	1.000	0.571	0.000	0.000
1374	adversarial.compensated.violation	Old audit	14	0.783	1.000	0.643	0.000	0.000
1375	adversarial.compensated.violation	Old audit	14	0.833	1.000	0.714	0.000	0.000
1376	adversarial.compensated.violation	Old audit	14	0.526	1.000	0.357	0.000	0.000
1377	adversarial.compensated.violation	Old audit	14	0.727	1.000	0.571	0.000	0.000
1378	adversarial.compensated.violation	Old audit	14	0.880	1.000	0.786	0.000	0.000
1379	adversarial.compensated.violation	Audit v5	14	0.526	1.000	0.357	0.000	0.000
1380	adversarial.compensated.violation	Audit v5	14	0.833	1.000	0.714	0.000	0.000
1381	adversarial.compensated.violation	Audit v5	14	0.727	1.000	0.571	0.000	0.000
1382	adversarial.compensated.violation	Audit v5	14	0.783	1.000	0.643	0.000	0.000
1383	adversarial.compensated.violation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1384	adversarial.compensated.violation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1385	adversarial.compensated.violation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1386	adversarial.compensated.violation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1387	adversarial.compensated.violation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1388	combined.violation.shift	Random	14	0.526	1.000	0.357	0.000	0.000
1389	combined.violation.shift	Random	14	0.667	1.000	0.500	0.000	0.000
1390	combined.violation.shift	Random	14	0.600	1.000	0.429	0.000	0.000
1391	combined.violation.shift	Random	14	0.667	1.000	0.500	0.000	0.000
1392	combined.violation.shift	Random	14	0.600	1.000	0.429	0.000	0.000
1393	combined.violation.shift	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1394	combined.violation.shift	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1395	combined.violation.shift	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1396	combined.violation.shift	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1397	combined.violation.shift	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1398	combined.violation.shift	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1399	combined.violation.shift	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1400	combined.violation.shift	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1401	combined.violation.shift	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1402	combined.violation.shift	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1403	combined.violation.shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000
	combined.violation.shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1404								
1405								
1406	combined_violation_shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1407	combined_violation_shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1408	combined_violation_shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1409	combined_violation_shift	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1410	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1411	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1412	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1413	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1414	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1415	combined_violation_shift	Max residual	14	1.000	1.000	1.000	0.000	0.000
1416	combined_violation_shift	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1417	combined_violation_shift	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1418	combined_violation_shift	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1419	combined_violation_shift	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1420	combined_violation_shift	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1421	combined_violation_shift	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1422	combined_violation_shift	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1423	combined_violation_shift	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1424	combined_violation_shift	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1425	combined_violation_shift	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1426	combined_violation_shift	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1427	combined_violation_shift	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1428	combined_violation_shift	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1429	combined_violation_shift	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1430	combined_violation_shift	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1431	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1432	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1433	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1434	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1435	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1436	combined_violation_shift	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1437	combined_violation_shift	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1438	combined_violation_shift	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1439	combined_violation_shift	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1440	combined_violation_shift	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1441	combined_violation_shift	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1442	combined_violation_shift	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1443	combined_violation_shift	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1444	combined_violation_shift	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1445	combined_violation_shift	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1446	combined_violation_shift	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1447	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1448	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1449	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1450	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1451	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1452	combined_violation_shift	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1453	combined_violation_shift	Old audit	14	1.000	1.000	1.000	0.000	0.000
1454	combined_violation_shift	Old audit	14	1.000	1.000	1.000	0.000	0.000
1455	combined_violation_shift	Old audit	14	1.000	1.000	1.000	0.000	0.000
1456	combined_violation_shift	Old audit	14	1.000	1.000	1.000	0.000	0.000
1457	combined_violation_shift	Old audit	14	1.000	1.000	1.000	0.000	0.000
		Audit v5	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1458								
1459	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1460	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1461	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1462	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1463	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1464	combined_violation_shift	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1465	combined_violation_shift	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1466	combined_violation_shift	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1467	combined_violation_shift	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1468	combined_violation_shift	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1469	combined_violation_shift	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1470	contact_corruption	Random	14	0.923	1.000	0.857	0.000	0.000
1471	contact_corruption	Random	14	0.727	1.000	0.571	0.000	0.000
1472	contact_corruption	Random	14	0.600	1.000	0.429	0.000	0.000
1473	contact_corruption	Random	14	0.833	1.000	0.714	0.000	0.000
1474	contact_corruption	Random	14	0.667	1.000	0.500	0.000	0.000
1475	contact_corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1476	contact_corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1477	contact_corruption	Random	14	0.727	1.000	0.571	0.000	0.000
1478	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1479	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1480	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1481	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1482	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1483	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1484	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1485	contact_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1486	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1487	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1488	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1489	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1490	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1491	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1492	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1493	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1494	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1495	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1496	contact_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1497	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1498	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1499	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1500	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1501	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1502	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1503	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1504	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1505	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1506	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1507	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1508	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1509	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1510	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000
1511	contact_corruption	Contact residual	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1512								
1513	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1514	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1515	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1516	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1517	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1518	contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
1519	contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1520	contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1521	contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1522	contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1523	contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1524	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1525	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1526	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1527	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1528	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1529	contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1530	contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1531	contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1532	contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1533	contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1534	contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1535	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1536	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1537	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1538	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1539	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1540	contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
1541	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1542	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1543	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1544	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1545	contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1546	contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1547	contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1548	contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1549	contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1550	contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1551	energy_work_corruption	Random	14	0.667	1.000	0.500	0.000	0.000
1552	energy_work_corruption	Random	14	0.727	1.000	0.571	0.000	0.000
1553	energy_work_corruption	Random	14	0.727	1.000	0.571	0.000	0.000
1554	energy_work_corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1555	energy_work_corruption	Random	14	0.667	1.000	0.500	0.000	0.000
1556	energy_work_corruption	Random	14	0.667	1.000	0.500	0.000	0.000
1557	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1558	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1559	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1560	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1561	energy_work_corruption	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1562	energy_work_corruption	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1563	energy_work_corruption	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1564	energy_work_corruption	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1565	energy_work_corruption	Energy residual	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1566								
1567	energy_work_corruption	Energy residual	14	1.000	1.000	1.000	0.000	0.000
1568	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1569	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1570	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1571	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1572	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1573	energy_work_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1574	energy_work_corruption	Max residual	14	0.526	1.000	0.357	0.000	0.000
1575	energy_work_corruption	Max residual	14	0.353	1.000	0.214	0.000	0.000
1576	energy_work_corruption	Max residual	14	0.133	1.000	0.071	0.000	0.000
1577	energy_work_corruption	Max residual	14	0.133	1.000	0.071	0.000	0.000
1578	energy_work_corruption	Max residual	14	0.600	1.000	0.429	0.000	0.000
1579	energy_work_corruption	Max residual	14	0.526	1.000	0.357	0.000	0.000
1580	energy_work_corruption	Max residual	14	0.133	1.000	0.071	0.000	0.000
1581	energy_work_corruption	Max residual	14	0.353	1.000	0.214	0.000	0.000
1582	energy_work_corruption	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
1583	energy_work_corruption	Logistic stack	14	0.963	1.000	0.929	0.000	0.000
1584	energy_work_corruption	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1585	energy_work_corruption	Logistic stack	14	0.963	1.000	0.929	0.000	0.000
1586	energy_work_corruption	Logistic stack	14	0.833	1.000	0.714	0.000	0.000
1587	energy_work_corruption	Logistic stack	14	0.963	1.000	0.929	0.000	0.000
1588	energy_work_corruption	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
1589	energy_work_corruption	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
1590	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1591	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1592	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1593	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1594	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1595	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1596	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1597	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1598	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1599	energy_work_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
1600	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1601	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1602	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1603	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1604	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1605	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1606	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1607	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1608	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1609	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1610	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1611	energy_work_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1612	energy_work_corruption	RF ensemble	14	1.000	1.000	1		

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1620								
1621	energy_work_corruption	Old audit	14	0.133	1.000	0.071	0.000	0.000
1622	energy_work_corruption	Old audit	14	0.353	1.000	0.214	0.000	0.000
1623	energy_work_corruption	Audit v5	14	0.526	1.000	0.357	0.000	0.000
1624	energy_work_corruption	Audit v5	14	0.353	1.000	0.214	0.000	0.000
1625	energy_work_corruption	Audit v5	14	0.133	1.000	0.071	0.000	0.000
1626	energy_work_corruption	Audit v5	14	0.600	1.000	0.429	0.000	0.000
1627	energy_work_corruption	Audit v5	14	0.526	1.000	0.357	0.000	0.000
1628	energy_work_corruption	Audit v5	14	0.133	1.000	0.071	0.000	0.000
1629	energy_work_corruption	Audit v5	14	0.353	1.000	0.214	0.000	0.000
1630	energy_work_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1631	energy_work_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1632	energy_work_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1633	energy_work_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1634	mixed_near_threshold	Random	14	0.444	1.000	0.286	0.000	0.000
1635	mixed_near_threshold	Random	14	0.667	1.000	0.500	0.000	0.000
1636	mixed_near_threshold	Random	14	0.783	1.000	0.643	0.000	0.000
1637	mixed_near_threshold	Random	14	0.727	1.000	0.571	0.000	0.000
1638	mixed_near_threshold	Random	14	0.353	1.000	0.214	0.000	0.000
1639	mixed_near_threshold	Random	14	0.727	1.000	0.571	0.000	0.000
1640	mixed_near_threshold	Kin. residual	14	0.880	1.000	0.786	0.000	0.000
1641	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1642	mixed_near_threshold	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1643	mixed_near_threshold	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
1644	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1645	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1646	mixed_near_threshold	Kin. residual	14	0.880	1.000	0.786	0.000	0.000
1647	mixed_near_threshold	Kin. residual	14	0.963	1.000	0.929	0.000	0.000
1648	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1649	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1650	mixed_near_threshold	Energy residual	14	0.133	1.000	0.071	0.000	0.000
1651	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1652	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1653	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1654	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1655	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1656	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1657	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1658	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1659	mixed_near_threshold	Energy residual	14	0.000	0.000	0.000	0.000	0.000
1660	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1661	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1662	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1663	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1664	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1665	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1666	mixed_near_threshold	Contact residual	14	0.000	0.000	0.000	0.000	0.000

[illegible]

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1836								
1837	noncausal.teleport	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1838	noncausal.teleport	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1839	noncausal.teleport	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1840	noncausal.teleport	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1841	noncausal.teleport	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
1842	noncausal.teleport	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1843	noncausal.teleport	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1844	noncausal.teleport	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1845	noncausal.teleport	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1846	noncausal.teleport	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
1847	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1848	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1849	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1850	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1851	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1852	noncausal.teleport	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
1853	noncausal.teleport	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1854	noncausal.teleport	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1855	noncausal.teleport	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1856	noncausal.teleport	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1857	noncausal.teleport	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
1858	noncausal.teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1859	noncausal.teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1860	noncausal.teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1861	noncausal.teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1862	noncausal.teleport	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
1863	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1864	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1865	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1866	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1867	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1868	noncausal.teleport	Old audit	14	1.000	1.000	1.000	0.000	0.000
1869	noncausal.teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1870	noncausal.teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1871	noncausal.teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1872	noncausal.teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1873	noncausal.teleport	Audit v5	14	1.000	1.000	1.000	0.000	0.000
1874	noncausal.teleport	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1875	noncausal.teleport	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1876	noncausal.teleport	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1877	noncausal.teleport	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1878	noncausal.teleport	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
1879	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.357	0.000
1880	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.500	0.000
1881	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.357	0.000
1882	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.357	0.000
1883	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.571	0.000
1884	rare_valid.bounce	Random	14	0.000	0.000	0.000	0.643	0.000
1885	rare_valid.bounce	Kin. residual	14	0.000	0.000	0.000	0.500	0.000
1886	rare_valid.bounce	Kin. residual	14	0.000	0.000	0.000	0.357	0.000
1887	rare_valid.bounce	Kin. residual	14	0.000	0.000	0.000	0.214	0.000
1888	rare_valid.bounce	Kin. residual	14	0.000	0.000	0.000	0.143	0.000
1889	rare_valid.bounce	Kin. residual	14	0.000	0.000	0.000	0.214	0.000
	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.214	0.000
	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.286	0.000
	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.643	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1890								
1891	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.143	0.000
1892	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.143	0.000
1893	rare_valid.bounce	HGB classifier	14	0.000	0.000	0.000	0.286	0.000
1894	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.143	0.000
1895	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.286	0.000
1896	rare_valid.bounce	Energy residual	14	0.000	0.000	0.000	0.286	0.000
1897	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.000	0.000
1898	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.143	0.000
1899	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.286	0.000
1900	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.071	0.000
1901	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.071	0.000
1902	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.214	0.000
1903	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.143	0.000
1904	rare_valid.bounce	Contact residual	14	0.000	0.000	0.000	0.214	0.000
1905	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.000	0.000
1906	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.000	0.000
1907	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1908	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.000	0.000
1909	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.286	0.000
1910	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.071	0.000
1911	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.071	0.000
1912	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.214	0.000
1913	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.071	0.000
1914	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.071	0.000
1915	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.214	0.000
1916	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1917	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1918	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1919	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1920	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1921	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1922	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1923	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.214	0.000
1924	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.286	0.000
1925	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.000	0.000
1926	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1927	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1928	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1929	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1930	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1931	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1932	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1933	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1934	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1935	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1936	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1937	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1938	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1939	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1940	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1941	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1942	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
1943	rare_valid.bounce	Max residual	14	0.000	0.000	0.000	0.143	0.000
	rare_valid.bounce	Calibrated stack	14	0.000	0.000	0.000	0.286	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1944								
1945	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.000	0.000
1946	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.000	0.000
1947	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.143	0.000
1948	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.000	0.000
1949	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.286	0.000
1950	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.071	0.000
1951	rare_valid.bounce	Old audit	14	0.000	0.000	0.000	0.071	0.000
1952	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1953	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.143	0.000
1954	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.000	0.000
1955	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.286	0.000
1956	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.071	0.000
1957	rare_valid.bounce	Audit v5	14	0.000	0.000	0.000	0.071	0.000
1958	rare_valid.bounce	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
1959	rare_valid.bounce	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
1960	rare_valid.bounce	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
1961	rare_valid.bounce	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
1962	subtle_contact.corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1963	subtle_contact.corruption	Random	14	0.667	1.000	0.500	0.000	0.000
1964	subtle_contact.corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1965	subtle_contact.corruption	Random	14	0.444	1.000	0.286	0.000	0.000
1966	subtle_contact.corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1967	subtle_contact.corruption	Random	14	0.727	1.000	0.571	0.000	0.000
1968	subtle_contact.corruption	Random	14	0.783	1.000	0.643	0.000	0.000
1969	subtle_contact.corruption	Kin. residual	14	0.833	1.000	0.714	0.000	0.000
1970	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1971	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1972	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1973	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1974	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1975	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1976	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1977	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1978	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1979	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1980	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1981	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1982	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1983	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1984	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1985	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1986	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1987	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1988	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1989	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1990	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1991	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1992	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1993	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1994	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1995	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1996	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
1997	subtle_contact.corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
1998								
1999								
2000	subtle_contact_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2001	subtle_contact_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2002	subtle_contact_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2003	subtle_contact_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2004	subtle_contact_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2005	subtle_contact_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2006	subtle_contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
2007	subtle_contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
2008	subtle_contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
2009	subtle_contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
2010	subtle_contact_corruption	IsolationForest	14	1.000	1.000	1.000	0.000	0.000
2011	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2012	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2013	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2014	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2015	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2016	subtle_contact_corruption	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2017	subtle_contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2018	subtle_contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2019	subtle_contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2020	subtle_contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2021	subtle_contact_corruption	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2022	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2023	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2024	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2025	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2026	subtle_contact_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2027	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2028	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2029	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2030	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2031	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2032	subtle_contact_corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2033	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2034	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2035	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2036	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2037	subtle_contact_corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2038	subtle_contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2039	subtle_contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2040	subtle_contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2041	subtle_contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2042	subtle_contact_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2043	subtle_energy_corruption	Random	14	0.833	1.000	0.714	0.000	0.000
2044	subtle_energy_corruption	Random	14	0.600	1.000	0.429	0.000	0.000
2045	subtle_energy_corruption	Random	14	0.783	1.000	0.643	0.000	0.000
2046	subtle_energy_corruption	Random	14	0.444	1.000	0.286	0.000	0.000
2047	subtle_energy_corruption	Random	14	0.833	1.000	0.714	0.000	0.000
2048	subtle_energy_corruption	Random	14	0.833	1.000	0.714	0.000	0.000
2049	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2050	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2051	subtle_energy_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2106								
2107								
2108	subtle.energy.corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2109	subtle.energy.corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2110	subtle.energy.corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2111	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2112	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2113	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2114	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2115	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2116	subtle.energy.corruption	Old audit	14	0.000	0.000	0.000	0.000	0.000
2117	subtle.energy.corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2118	subtle.energy.corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2119	subtle.energy.corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2120	subtle.energy.corruption	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2121	subtle.energy.corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2122	subtle.energy.corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2123	subtle.energy.corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2124	subtle.energy.corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2125	subtle.energy.corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2126	support.levitation	Random	14	0.727	1.000	0.571	0.000	0.000
2127	support.levitation	Random	14	0.727	1.000	0.571	0.000	0.000
2128	support.levitation	Random	14	0.880	1.000	0.786	0.000	0.000
2129	support.levitation	Random	14	0.526	1.000	0.357	0.000	0.000
2130	support.levitation	Random	14	0.727	1.000	0.571	0.000	0.000
2131	support.levitation	Random	14	0.727	1.000	0.571	0.000	0.000
2132	support.levitation	Random	14	0.727	1.000	0.571	0.000	0.000
2133	support.levitation	Random	14	0.353	1.000	0.214	0.000	0.000
2134	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2135	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2136	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2137	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2138	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2139	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2140	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2141	support.levitation	Kin. residual	14	1.000	1.000	1.000	0.000	0.000
2142	support.levitation	Energy residual	14	0.600	1.000	0.429	0.000	0.000
2143	support.levitation	Energy residual	14	0.353	1.000	0.214	0.000	0.000
2144	support.levitation	Energy residual	14	0.250	1.000	0.143	0.000	0.000
2145	support.levitation	Energy residual	14	0.444	1.000	0.286	0.000	0.000
2146	support.levitation	Energy residual	14	0.667	1.000	0.500	0.000	0.000
2147	support.levitation	Energy residual	14	0.526	1.000	0.357	0.000	0.000
2148	support.levitation	Energy residual	14	0.353	1.000	0.214	0.000	0.000
2149	support.levitation	Energy residual	14	0.727	1.000	0.571	0.000	0.000
2150	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2151	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2152	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2153	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2154	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2155	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2156	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2157	support.levitation	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2158	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
2159	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Max residual	14	0.000	0.000	0.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	Logistic stack	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
	support.levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2160								
2161	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2162	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2163	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2164	support_levitation	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2165	support_levitation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2166	support_levitation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2167	support_levitation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2168	support_levitation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2169	support_levitation	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2170	support_levitation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2171	support_levitation	IsolationForest	14	0.133	1.000	0.071	0.000	0.000
2172	support_levitation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2173	support_levitation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2174	support_levitation	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2175	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2176	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2177	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2178	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2179	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2180	support_levitation	PCA recon.	14	1.000	1.000	1.000	0.000	0.000
2181	support_levitation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2182	support_levitation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2183	support_levitation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2184	support_levitation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2185	support_levitation	Conformal residual	14	1.000	1.000	1.000	0.000	0.000
2186	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2187	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2188	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2189	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2190	support_levitation	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2191	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2192	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2193	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2194	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2195	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2196	support_levitation	Old audit	14	0.000	0.000	0.000	0.000	0.000
2197	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2198	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2199	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2200	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2201	support_levitation	Audit v5	14	0.000	0.000	0.000	0.000	0.000
2202	support_levitation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2203	support_levitation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2204	support_levitation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2205	support_levitation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2206	support_levitation	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2207	timestamp_noncausal_corruption	Random	14	0.833	1.000	0.714	0.000	0.000
2208	timestamp_noncausal_corruption	Random	14	0.600	1.000	0.429	0.000	0.000
2209	timestamp_noncausal_corruption	Random	14	0.667	1.000	0.500	0.000	0.000
2210	timestamp_noncausal_corruption	Random	14	0.783	1.000	0.643	0.000	0.000
2211	timestamp_noncausal_corruption	Random	14	0.526	1.000	0.357	0.000	0.000
2212	timestamp_noncausal_corruption	Random	14	0.727	1.000	0.571	0.000	0.000
2213	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
	timestamp_noncausal_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2214								
2215	timestamp_noncausal_corruption	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2216	timestamp_noncausal_corruption	Kin. residual	14	0.250	1.000	0.143	0.000	0.000
2217	timestamp_noncausal_corruption	Kin. residual	14	0.133	1.000	0.071	0.000	0.000
2218	timestamp_noncausal_corruption	Kin. residual	14	0.133	1.000	0.071	0.000	0.000
2219	timestamp_noncausal_corruption	Kin. residual	14	0.353	1.000	0.214	0.000	0.000
2220	timestamp_noncausal_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2221	timestamp_noncausal_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2222	timestamp_noncausal_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2223	timestamp_noncausal_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2224	timestamp_noncausal_corruption	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2225	timestamp_noncausal_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2226	timestamp_noncausal_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2227	timestamp_noncausal_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2228	timestamp_noncausal_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2229	timestamp_noncausal_corruption	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2230	timestamp_noncausal_corruption	Max residual	14	0.667	1.000	0.500	0.000	0.000
2231	timestamp_noncausal_corruption	Max residual	14	0.444	1.000	0.286	0.000	0.000
2232	timestamp_noncausal_corruption	Max residual	14	0.600	1.000	0.429	0.000	0.000
2233	timestamp_noncausal_corruption	Max residual	14	0.727	1.000	0.571	0.000	0.000
2234	timestamp_noncausal_corruption	Max residual	14	0.833	1.000	0.714	0.000	0.000
2235	timestamp_noncausal_corruption	Max residual	14	0.727	1.000	0.571	0.000	0.000
2236	timestamp_noncausal_corruption	Max residual	14	0.833	1.000	0.714	0.000	0.000
2237	timestamp_noncausal_corruption	Logistic stack	14	0.783	1.000	0.643	0.000	0.000
2238	timestamp_noncausal_corruption	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
2239	timestamp_noncausal_corruption	Logistic stack	14	0.923	1.000	0.857	0.000	0.000
2240	timestamp_noncausal_corruption	Logistic stack	14	0.783	1.000	0.643	0.000	0.000
2241	timestamp_noncausal_corruption	Logistic stack	14	0.727	1.000	0.571	0.000	0.000
2242	timestamp_noncausal_corruption	Logistic stack	14	0.667	1.000	0.500	0.000	0.000
2243	timestamp_noncausal_corruption	Logistic stack	14	0.833	1.000	0.714	0.000	0.000
2244	timestamp_noncausal_corruption	Logistic stack	14	0.880	1.000	0.786	0.000	0.000
2245	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2246	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2247	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2248	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2249	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2250	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2251	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2252	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2253	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2254	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2255	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2256	timestamp_noncausal_corruption	HGB classifier	14	1.000	1.000	1.000	0.000	0.000
2257	timestamp_noncausal_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2258	timestamp_noncausal_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2259	timestamp_noncausal_corruption	RF ensemble	14	1.000	1.000	1.000	0.000	0.000
2260	timestamp_noncausal_corruption							

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2268								
2269	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2270	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2271	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2272	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2273	timestamp_noncausal_corruption	Calibrated stack	14	1.000	1.000	1.000	0.000	0.000
2274	timestamp_noncausal_corruption	Old audit	14	0.667	1.000	0.500	0.000	0.000
2275	timestamp_noncausal_corruption	Old audit	14	0.667	1.000	0.500	0.000	0.000
2276	timestamp_noncausal_corruption	Old audit	14	0.444	1.000	0.286	0.000	0.000
2277	timestamp_noncausal_corruption	Old audit	14	0.600	1.000	0.429	0.000	0.000
2278	timestamp_noncausal_corruption	Old audit	14	0.727	1.000	0.571	0.000	0.000
2279	timestamp_noncausal_corruption	Old audit	14	0.833	1.000	0.714	0.000	0.000
2280	timestamp_noncausal_corruption	Audit v5	14	0.667	1.000	0.500	0.000	0.000
2281	timestamp_noncausal_corruption	Audit v5	14	0.667	1.000	0.500	0.000	0.000
2282	timestamp_noncausal_corruption	Audit v5	14	0.444	1.000	0.286	0.000	0.000
2283	timestamp_noncausal_corruption	Audit v5	14	0.600	1.000	0.429	0.000	0.000
2284	timestamp_noncausal_corruption	Audit v5	14	0.727	1.000	0.571	0.000	0.000
2285	timestamp_noncausal_corruption	Audit v5	14	0.833	1.000	0.714	0.000	0.000
2286	timestamp_noncausal_corruption	Audit v5	14	0.727	1.000	0.571	0.000	0.000
2287	timestamp_noncausal_corruption	Audit v5	14	0.783	1.000	0.643	0.000	0.000
2288	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2289	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2290	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2291	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2292	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2293	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2294	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2295	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2296	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2297	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2298	timestamp_noncausal_corruption	Oracle labels	14	1.000	1.000	1.000	0.000	0.000
2299	valid_clock_skew	Random	14	0.000	0.000	0.000	0.286	0.000
2300	valid_clock_skew	Random	14	0.000	0.000	0.000	0.286	0.000
2301	valid_clock_skew	Random	14	0.000	0.000	0.000	0.714	0.000
2302	valid_clock_skew	Random	14	0.000	0.000	0.000	0.571	0.000
2303	valid_clock_skew	Random	14	0.000	0.000	0.000	0.571	0.000
2304	valid_clock_skew	Random	14	0.000	0.000	0.000	0.571	0.000
2305	valid_clock_skew	Random	14	0.000	0.000	0.000	0.500	0.000
2306	valid_clock_skew	Random	14	0.000	0.000	0.000	0.214	0.000
2307	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2308	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2309	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2310	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2311	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2312	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2313	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2314	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2315	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2316	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2317	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2318	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2319	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2320	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2321	valid_clock_skew	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.214	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2322								
2323	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
2324	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2325	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
2326	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.143	0.000
2327	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.071	0.000
2328	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.214	0.000
2329	valid_clock_skew	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2330	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.214	0.000
2331	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.214	0.000
2332	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2333	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2334	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.143	0.000
2335	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.143	0.000
2336	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2337	valid_clock_skew	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2338	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2339	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2340	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2341	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2342	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2343	valid_clock_skew	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2344	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2345	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2346	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2347	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2348	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2349	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.071	0.000
2350	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2351	valid_clock_skew	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2352	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2353	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2354	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2355	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2356	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2357	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2358	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2359	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2360	valid_clock_skew	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2361	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.143	0.000
2362	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.214	0.000
2363	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2364	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2365	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.071	0.000
2366	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2367	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2368	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2369	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2370	valid_clock_skew	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
2371	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.571	0.000
2372	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.571	0.000
2373	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.571	0.000
2374	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.429	0.000
2375	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.500	0.000
	valid_low_friction_slip	Random	14	0.000	0.000	0.000	0.429	0.000

	Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
2376								
2377								
2378	valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2379	valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2380	valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2381	valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2382	valid_low_friction_slip	Kin. residual	14	0.000	0.000	0.000	0.000	0.000
2383	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2384	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2385	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2386	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2387	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2388	valid_low_friction_slip	Energy residual	14	0.000	0.000	0.000	0.000	0.000
2389	valid_low_friction_slip	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2390	valid_low_friction_slip	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2391	valid_low_friction_slip	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2392	valid_low_friction_slip	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2393	valid_low_friction_slip	Contact residual	14	0.000	0.000	0.000	0.000	0.000
2394	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.214	0.000
2395	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.143	0.000
2396	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.214	0.000
2397	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.357	0.000
2398	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.143	0.000
2399	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.500	0.000
2400	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.429	0.000
2401	valid_low_friction_slip	Max residual	14	0.000	0.000	0.000	0.286	0.000
2402	valid_low_friction_slip	Logistic stack	14	0.000	0.000	0.000	0.000	0.000
2403	valid_low_friction_slip	Logistic stack	14	0.000	0.000	0.000	0.000	0.000
2404	valid_low_friction_slip	Logistic stack	14	0.000	0.000	0.000	0.000	0.000
2405	valid_low_friction_slip	Logistic stack	14	0.000	0.000	0.000	0.000	0.000
2406	valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2407	valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2408	valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2409	valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2410	valid_low_friction_slip	HGB classifier	14	0.000	0.000	0.000	0.000	0.000
2411	valid_low_friction_slip	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2412	valid_low_friction_slip	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2413	valid_low_friction_slip	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2414	valid_low_friction_slip	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2415	valid_low_friction_slip	RF ensemble	14	0.000	0.000	0.000	0.000	0.000
2416	valid_low_friction_slip	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2417	valid_low_friction_slip	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2418	valid_low_friction_slip	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2419	valid_low_friction_slip	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2420	valid_low_friction_slip	IsolationForest	14	0.000	0.000	0.000	0.000	0.000
2421	valid_low_friction_slip	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2422	valid_low_friction_slip	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2423	valid_low_friction_slip	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2424	valid_low_friction_slip	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2425	valid_low_friction_slip	PCA recon.	14	0.000	0.000	0.000	0.000	0.000
2426	valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2427	valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
2428	valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.071	0.000
2429	valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
	valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.000	0.000

Split	Method	Episodes	F1	Prec.	Rec.	FPR	AUROC
valid_low_friction_slip	Conformal residual	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Calibrated stack	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.357	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.500	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.429	0.000
valid_low_friction_slip	Old audit	14	0.000	0.000	0.000	0.286	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.214	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.357	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.143	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.500	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.429	0.000
valid_low_friction_slip	Audit v5	14	0.000	0.000	0.000	0.286	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000
valid_low_friction_slip	Oracle labels	14	0.000	0.000	0.000	0.000	0.000

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